Multistage Superpixel-Guided Hyperspectral Image Classification With Sparse Graph Attention Networks

Weiming Li, *Member, IEEE*, Qikang Liu, Shuaishuai Fan, *Member, IEEE*, Hongyang Bai, *Member, IEEE*, and Mingrui Xin, *Member, IEEE*

***Abstract*— Graph neural networks (GNNs), as an effective learning framework for graph structure data representation, have been applied to hyperspectral images (HSIs) classification tasks. Among the variants of GNNs, graph attention networks (GATs) have achieved state-of-the-art node prediction performance by learning to assign dense attention coefficients to all node neighbors for feature aggregation. However, due to the complexity of land distribution and the high dimension of HSIs data, it is difficult to identify different coverage categories by employing GATs directly, and the inadequacy of spectral information also affects the classification accuracy. To address the above application problems, we proposed a multistage superpixel-guided sparse GAT (MSG-SGAT) for HSI classification. Specifically, we create the adjacency connection graphs of different stages from the superpixel representation, so as to effectively utilize the spatial topology. An SGAT module is designed to trim the graph by spectral sparsity to remove some task-irrelevant edges and assign a unique attention coefficient to each remaining edge. Sparse subgraphs are obtained by identifying the noise/task-independent edges and aggregating information. Moreover, SGAT is concatenated with the spectral branch for feature fusion and update, and pixel-level feature refinement is performed at the end of this network. MSG-SGAT mines the features of HSIs from the perspective of a multiscale hierarchy with low calculation cost and high efficiency. A series of comparative experiments on four benchmark datasets demonstrate that the performance of our proposed method is competitive with others.**

***Index Terms*— Graph convolution, hyperspectral image (HSI) classification, spectral sparsity, superpixel segmentation.**

I. Introduction

HYPERSPECTRAL images (HSIs) can record spatial and spectral information simultaneously with strong correlations between the various bands. The spectral features of HSIs contain various categories of land cover information, and the category affiliation is determined according to the spectral reflection value of pixels in each band. As the fundamental research of remote sensing application, HSI classification task is widely used in vegetation monitoring, resource exploration, environmental monitoring, and other fields [1], [2], [3], [4], [5]. However, due to some factors such as the dark current in photodetectors, atmospheric mixing effect, and the diversity of ground objects, a significant amount of noise is introduced into the imaging data, which makes a certain impact on HSI classification task. Therefore, how to extract deep-level and strong anti-interference features during the classification process has great practical value.

Although the early statistical methods, such as decision tree [6], random forest [7], and support vector machines (SVMs) [8], have improved the classification effect of HSI to some extent, they have not overcome the limitations posed by the high dimensionality of HSI and the nonlinear separability of pixels. In order to further reveal the correlation relationship between the spatial and spectral domains in HSI, techniques such as constrained energy minimization (CEM), active learning (AL), and iterative training sampling (ITS) are also applied to HSI classification. Chang and Ma [9] exploit the subpixel detection capability of CEM to develop a nonlinear hybrid pixel classifier and incorporate band sampling into the Nystrom method-based kernel CEM to form two new versions of hyperspectral hybrid pixel classification. In addition, a joint spectral–spatial classification method named ITSA-AL-SS [10] is also proposed to adopt ITS for extending the data cube by iteratively merging additional spatial information into AL-enhanced unlabeled data samples to improve the classification performance. These methods have been successfully applied in the field of hyperspectral classification and have shown good performance. However, the above methods have a common problem that limits their application scope to some extent, that is, the original spatial information is lost in the process of dimension reduction, and the complex spectral spatial structure and nonlinear relationship cannot be captured.

Deep learning methods, typically represented by convolutional neural networks (CNNs), have shown great potential for HSI classification in extracting hierarchical and high-level semantic features, which can effectively reduce the possibility of information loss caused by dimensionality reduction in the preprocess. CNN-based methods can be divided into

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three categories according to the type of convolution kernels. 1-D-CNN-based methods, such as multilayer perceptron [11], autoencoder [12], and the self-organizing mapping neural network [13], are often used as feature extractors to collect discriminative spectral features. However, the input tensors must be flattened to adapt to the computational form, which ignores the spatial context of the original HSI pixels. In contrast, 2-D-CNN-based methods with diverse architectures, convolution kernels, and connection relations have been well applied in HSI classification. For instance, Gao et al. [14] propose a novel CNN with multifeature learning, which performs HSI pixel-level classification by fusing multiple features. Li et al. [15] present a two-branch dual-attention mechanism network that captures the spectral and spatial features to optimize the extracted features. Wang et al. [16] propose an end-to-end fast dense spectral–spatial network that adopts multisize convolution kernels and dense connection to achieve accurate classification. However, the aforementioned methods still have certain limitations, that is, the high dimensionality of HSI poses challenges in simultaneously extracting spectral and spatial features, which also cause some disruption to the 3-D structure of HSI data. Therefore, some 3-D-CNN-based methods [17], [18], [19] have been proposed to solve the issues of gradient vanishing and information redundancy while preserving the 3-D data structure. In conclusion, although the aforementioned CNN-based HSI classification methods have achieved certain results, some application limitations still remain: 1) the spatial topology of actual land surface distribution is hard to characterize; 2) near-range spatial dependence in homogeneous regions is difficult to represent completely; 3) most models have a large number of parameters and low efficiency; and 4) overfitting is easy to occur due to the small number of available labeled samples.

Recently, graph neural networks (GNNs) have achieved excellent performance in various classification tasks by adopting the geometric structure of graphs to learn improved nodes and edge representation [20], [21], [22], [23], [24]. Since the unlabeled HSI data are more readily available, the semisupervised methods, such as GNNs, can potentially achieve efficient information aggregation using labeled and unlabeled samples. In variants of GNN, the graph attention network (GAT) effectively improves the performance of graph learning by generating dense attention coefficients among nodes, which are suitable for HSI classification tasks limited by data annotations. However, the existing GAT-based HSI classification methods still have two main problems.

1) *Huge Computation Overhead:* Although some methods introduce superpixel segmentation [25] to reduce the input dimension, in practice, GATs generate multiple attention coefficients for edges of each layer, which increases computational overhead and overfitting. Therefore, the key to employ GATs reasonably for HSI is to reduce input nodes and internal parameters, simultaneously.
2) The spatial–spectral interaction and the guiding role of multiscale information are ignored. Considering that actual land covers have different hierarchical tendencies in the feature scale, multiscale fusion guidance should be adopted to take into account the homogenous covers of different areas to improve model stability. In addition, the combination of spectral vectors and spatial details should be used to preserve actual topological relationships and contextual information.

To address the above issues, this article proposes a multistage superpixel-guided sparse GAT (MSG-SGAT) for HSI classification that is exhibited in Fig. 1. To homogenize local land covers and improve efficiency, the multistage superpixel segmentation mechanism based on region merging is adopted to perform hierarchical aggregation, which guides to construct multilevel graphs to comprehensively explore the spatial topology of HSIs. The core component of this network is employing sparse operations to design an SGAT to eliminate task-independent edges, which achieves parameters lightweight and performs adaptive updating of features at the spatial level. Our MSG-SGAT achieves state-of-the-art performance on four real hyperspectral benchmark datasets. The main contributions of this article are fourfold.

1) We propose a multistage graph structure network guided by superpixels for HSI classification, which is a U-shaped architecture with the encoder and the decoder. The adjacency connection graph of different stages is created from the superpixel representation to effectively utilize the spatial topology by connecting adjacent superpixels.
2) An SGAT is proposed to trim the graph through spectral sparsification to remove those edges irrelevant to the task and assign a unique attention coefficient to each remaining edge for scaling down parameters. SGAT simplifies the architecture of GATs and reduces the risk of overfitting while maintaining its accuracy advantage.
3) We design a spectral information acquisition (SIA) module to model the spectral relationship for effectively capturing the spectral features, which is connected with SGAT in parallel to form a spatial–spectral cross-unit (SSCU) with two branches to achieve information complementarity. Furthermore, the SSCU is merged with the upsampling operation to serve as the decoding unit, while it forms the encoding unit by adding downsampling operations.
4) A pixel-level transformation (PLT) module is designed to learn the conversion relationship from superpixels to pixels in order to refine and extract the final features. It is connected in series with the backbone to form an end-to-end network.

The reminder of this article is organized as follows. Section II presents the relevant work on variations of GAT and their applications. Section III introduces our MSG-SGAT and the design details of some modules. The focus is on implementing the SGAT module by using spectral sparsification for reducing the computation significantly while maintaining accuracy. Section IV reports the extensive experimental results and analysis. Section V concludes this article.

II. Related Work

In this section, we review some representative work that is closely related to this article, including variants of GAT and their applications.

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A. Variants

Recently, many variants of GATs have been proposed focusing on different directions, such as speed optimization [26], lightweight design [27], multilevel modeling [28], and multimodel integration [29]. These variants have been widely applied in various domains due to their respective advantages. In terms of reducing computational complexity, Wang et al. [30] proposed an edge-featured GAT that utilized both node and edge features so that the model could adaptively aggregate edges and nodes according to the varying preferences of graph features. Ye and Ji [31] introduced a regularized SGAT that used L0 regularization to learn sparse attention coefficients and form edge-sparse subgraphs. Chen et al. [32] designed a layerwise self-adaptive GAT, which took into account both adjacent and nonadjacent nodes from a global perspective, effectively alleviating the problem of oversmoothing in deep GAT. Ma et al. [33] embedded the relative position into the GAT to capture the structure and position information of nodes, thus enhancing the representation ability of features. Chang et al. [34] used weighted filters and graph wavelet bases for learning attention weights for different frequency components to form a spectral GAT network. Considering the utilization efficiency of spectral information in high-dimensional HSI data, we employ a spectral sparse method to prune the graph from the perspective of spectral feature extraction, aiming to reduce the size and computational cost of GAT.

B. Applications

GATs have attracted increasing attention due to their ability to handle irregular data, and their application in solving high-dimensional HSI data has also demonstrated superior performance. For instance, Wan et al. [35] proposed a multilevel graph network for HSI classification, which could automatically learn local and global contextual information among graph nodes to enhance the representation capability. GATs were also used in combination with other graph methods to obtain graph structure information more efficiently. Ding et al. [36] introduced a novel graph sampling and aggregation network that learned multiscale features from local area graphs and focused on important spatial targets through the GAT, allowing the network more targeted to obtain deep graph global information. Pu et al. [37] presented a GAT combined with local spectral filtering for HSI classification that used local spatial and spectral information efficiently to complement. However, the above GAT variants for HSI classification were all superpixel-level modeled, which could not extract fine features at the pixel level. To take advantage of GATs and combine them with the pixel-level feature extraction capability of CNNs, Wang et al. [38] presented a spectral pyramid GAT that used a 3-D-CNN to extract multiscale spectral information and then applied a GAT to explicitly extract higher order spatial features and improve accuracy. Different from the above existing variants, our proposed MSG-SGAT can exploit the multiscale HSI information and interactively acquire more discriminative features in spatial and spectral dimensions without aggregating irrelevant information during graph convolution. In addition, we perform pixel-level refinement on the rough features at the end of the backbone to enhance the overall classification performance of our network.

III. Proposed Method

In this section, the proposed methodology for addressing the GAT-based HSI classification problem is described. First, a brief explanation of the architecture is given. Then, the methods of constructing the correlation matrix and multilevel graph topology are represented after introducing the superpixel-guided merging strategy. Next, we formulate an SGAT methodology that uses spectral sparsification in the context of attention for node updating. Finally, the proposed SIA and PLT are described, respectively.

The whole architecture of our MSG-SGAT is illustrated in Fig. 1, which is similar to the U-Net [39] with two arms, i.e., the encoder and the decoder. The initial input to this network is an HSI denoted as $\boldsymbol{X} \in \mathbb{R}^{H\times W\times C}$ with height H, width W, and the number of bands C. We use superpixels as guidance to construct multistage graphs for obtaining the adjacency and incidence matrices with upsampling and downsampling functions. Two convolutional layers are adopted to squeeze and denoise the features of the original $\boldsymbol{X}$. In each stage, a crossing unit composed of the SGAT and SIA is employed to effectively extract the information of the spatial–spatial dimensions. The encoder is designed to extract features from the scene and expand the receptive field by changing the feature dimensions; symmetrically, the decoder is integrated into the same stages of the encoder by using a skip connection to result from the acquired properties. The features from the last stage of the decoder are refined to some extent even though they are propagated from the superpixel level to the pixel level. The PLT module is designed to further refine the features for mitigating the degradation of resolution caused by the coding process. Then, the SoftMax activation is adopted at the end to identify pixels and obtain the final output $\boldsymbol{Y} \in \mathbb{R}^{H\times W\times C'}$ with channel C'.

A. Superpixel-Guided Merging Strategy

Considering the inherent characteristics of HSIs, such as high dimensionality and correlation among bands, transforming original pixels as graph nodes in the graph convolution directly will produce a huge computational overhead and destroy the spatial topology. Furthermore, some sampling or statistical features can be adopted to replace land cover topotaxy and homogeneity in a certain scale region in HSIs since the surface distribution is regional. Therefore, aiming at the data dimension reduction and the stability elevation of our model, the superpixel segmentation algorithm via region merging is adopted to guide the generation of graph nodes and topology construction. Principal component bands should be selected to achieve dimensionality reduction previous to superpixel segmentation, and the potential impact of noises caused by detectors, atmosphere, illumination, other influences in HSIs, and redundant information should be removed simultaneously. Since the selected principal component bands should contain as much information as possible in the corresponding

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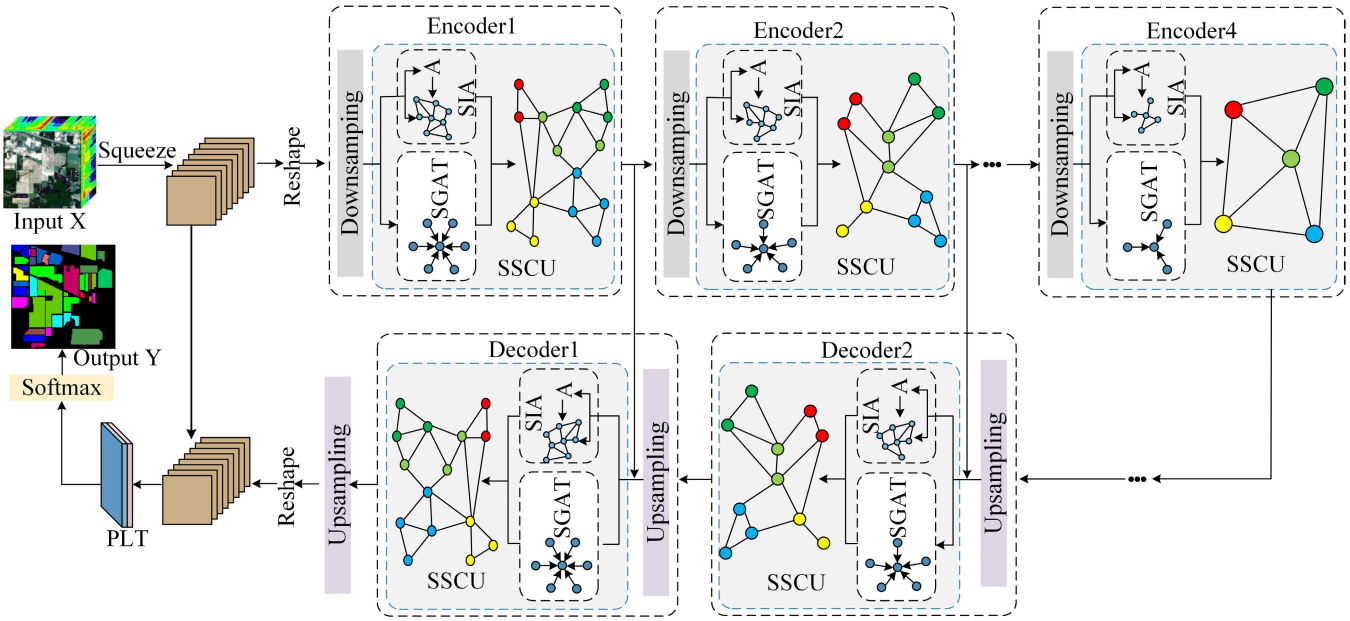


Fig. 1. Architecture of our proposed MSG-SGAT. $\boldsymbol{X}$ and $\boldsymbol{Y}$ denote the input data and output probability map of this network, respectively. The encoder and decoder parts are composed of SSCU in series, which contains the SGAT and SIA. The downsampling and upsampling operations are used to scale the features of the same stage in the two arms for concatenating channels.

datasets, we adopt the super hierarchy (SH) [40] to perform superpixel segmentation of the principal bands in HSIs, considering its simplicity, rapidity, and stability. It should be noted that other superpixel segmentation methods [41], [42], [43], [44] can also be appropriate for this article. Then, the key strategy is to aggregate adjacent homogeneous regions into superpixels via region merging, which serves as the guidance for constructing the graph topologies at different stages in the pipeline to exploit multiscale features of HSIs.

Specifically, the undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ of a given HSI is constructed with n nodes $v \in \mathcal{V}$, which are represented as the average of all pixels within each superpixel, and $e \in \mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ are defined as m edges of $\mathcal{G}$. Each node v is connected to its four-connected neighbors, and the weight w of each e is calculated by the spatial Euclidean distance between nodes. According to the principle of nearest neighborhood searching, v_j will be merged into v_i if w_{ij} is the smallest in the neighborhood of v_i. The nodes subset $\mathcal{V}_s$ ($\mathcal{V}_s \subseteq \mathcal{V}$) is obtained by repeated the merging strategy, where s denotes the number of remaining nodes, that is, a graph topology containing s superpixels is formed. Furthermore, a hierarchical architecture is built with generating any number of superpixels dynamically during this merging process, and we can create multilevel graphs with any number of nodes.

Let $\{\mathcal{S}^{(l)}, \mathcal{G}^{(l)}\}_{l=1}^{L}$ denote the lth superpixel segmentations and the corresponding graph topology sets, where $\mathcal{S}^{(l)} = \{s_i^{(l)}\}_{i=1}^{N}$ contains N superpixels at the lth layer and $\mathcal{G}^{(l)} = (\mathcal{V}^{(l)}, \mathcal{E}^{(l)}) = (\{v_i^{(l)}\}_{i=1}^{N}, \{e_j^{(l)}\}_{k=1}^{M})$ denotes the lth graph with N nodes and M edges. $\mathcal{G}^{(l)}$ is transformed from $\mathcal{S}^{(l)}$, and the transformation relationship between the two representations is shown in Fig. 2. Specifically, $\mathcal{V}^{(l)}$ and $\mathcal{E}^{(l)}$ are represented by a node matrix $\boldsymbol{H}^{(l)} \in \mathbb{R}^{N \times D}$ (D is the feature dimension) and an adjacency matrix $\boldsymbol{A}^{(l)} \in \mathbb{R}^{N \times N}$. Moreover, attention coefficient $a_{ij}^{(l)}$ of $\boldsymbol{A}^{(l)}$ is defined as the node features for every

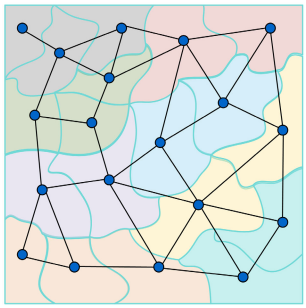


Fig. 2. Corresponding relations between superpixels and graph nodes.

pair of connected nodes $v_i, v_j \in \mathcal{V}^{(l)}$ at layer l. Essentially, the aggregator function aggregates low-level features of nodes and their neighbors to generate high-level feature representations. The attention-based aggregator can be expressed with $a_{ij}^{(l)}$, $h_i^{(l)}$ and the weight matrix $\boldsymbol{W}^{(l)}$.

Assuming that there are two superpixels adjacent to each other spatially at layer l, which are first-order neighbors with shared edges, then the element value at the corresponding position of $\boldsymbol{A}^{(l)}$ is 1, otherwise 0. The matrix $\boldsymbol{A}^{(l)}$ construction process is demonstrated in Fig. 3, where $a_{12}^{(l)} = a_{21}^{(l)} = 1$, $a_{13}^{(l)} = a_{31}^{(l)} = 1$, and $a_{23}^{(l)} = a_{32}^{(l)} = 1$, due to the spatial adjacency of regions 1, 2, and 3. For the sake of improving the adaptability of the graph, we employ the SGAT module (see Section II-B) to obtain node embedding and use $\boldsymbol{A}^{(l)}$ to update the nodes at layer l.

The mapping relationship of superpixels between two adjacent stages is constructed to achieve the progressive transfer of the graph between layers. The incidence matrix $\boldsymbol{C}^{(l)}$ is utilized for feature transformation in any two adjacent stages between the encoder and the decoder, which is described in pseudocode of Algorithm 1. w and h are the width and the

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Algorithm 1 Calculate the Correlation Matrix $\boldsymbol{C}$

Input: w, h, N, M, seg, $nextseg$, the values in seg and nextseg represent the superpixel labels for each pixel in the current and next layer images.

```
C[N, M] ← 0
for 1 ≤ x ≤ (w * h) do
    if C[seg[x], nextseg[x]] ≠ 1 then
        C[seg[x], nextseg[x]] = 1 //Correlation matrix between two adjacent layers
    end if
end for
return C
```

Output: $\boldsymbol{C}$

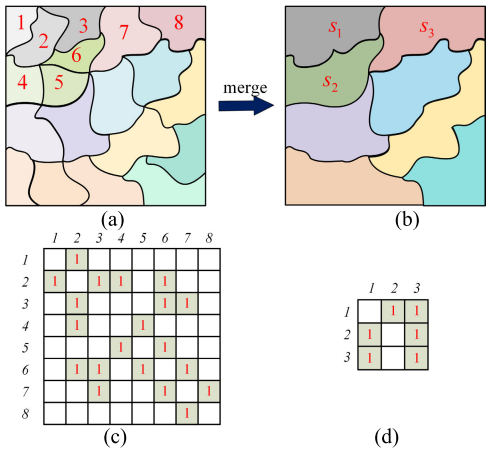


Fig. 3. Illustration of constructing multilevel features and adjacency matrices using merging strategy. (a) Superpixel segmentation at layer l, "1, 2, . . . , 8," is the serial number of the superpixel, respectively. (b) Merging the adjacent regions of (a) to generate superpixels at layer $l+1$, for instance, $\{1,2,3\} \rightarrow \{s_1\}$, $\{7,8\} \rightarrow \{s_3\}$. (c) Adjacency matrix of (a). (d) Adjacency matrix of (b).

height of the input feature, and N and M represent the number of superpixels in the current and next layers, respectively.

The node update process in the encoder can be written as

$$\boldsymbol{\mathcal{V}}^{(l+1)} = (\hat{\boldsymbol{C}}^{(l)})^{\mathrm{T}}\boldsymbol{\mathcal{V}}^{(l)} \leftarrow \mathrm{Encoder}(\boldsymbol{\mathcal{V}} \odot \boldsymbol{C}) \tag{1}$$

where $\hat{\boldsymbol{C}}^{(l)}$ represents the column-normalized incidence matrix of layer l and symbol $\odot$ indicates the inner product operation.

Equation (1) means that the superpixel numbered j at layer $l+1$ is generated by merging n superpixels from layer l, and its element calculation process is

$$\hat{c}_{ij}^{(l)} = \frac{c_{ij}^{(l)}}{\sum_{i=1}^{n} c_{ij}^{(l)}}. \tag{2}$$

Similarly, the transformation process of graph features from layer $l+1$ to l in the decoder can be represented as follows:

$$\tilde{\boldsymbol{\mathcal{V}}}^{(l)} = (\tilde{\boldsymbol{C}}^{(l)})^{\mathrm{T}}\boldsymbol{\mathcal{V}}^{(l+1)} \leftarrow \mathrm{Decoder}\big(\tilde{\boldsymbol{\mathcal{V}}} \odot \boldsymbol{C}\big). \tag{3}$$

In addition, $\tilde{\boldsymbol{C}}^{L}$ is defined as the row-normalized incidence matrix of layer l, and $\tilde{\boldsymbol{\mathcal{V}}}^{L+1}$ represents the nodes at layer $l+1$ in the decoder. In the decoding process, the superpixel numbered j at layer $l+1$ can be decomposed into m superpixels that belong to the lth layer. At the same stage of the two arms, m is equal to n. The features of different layers are concatenated through the U-shaped structure to carry out end-to-end training. In the encoder, the elements in the correlation matrix $\tilde{\boldsymbol{C}}^{L}$ are calculated as

$$\tilde{c}_{ij}^{(l)} = \frac{c_{ij}^{(l)}}{\sum_{j=1}^{m} c_{ij}^{(l)}} \tag{4}$$

where $c_{ij}^{(l)}$ denotes the element value in row i and column j, and $\tilde{c}_{ij}^{(l)}$ denotes the normalized $c_{ij}^{(l)}$.

B. SGAT Module

Using GAT to adaptively capture the correlation among superpixel centers in HSI is beneficial to maximize the aggregation of information within the same class, whereas the multihead attention mechanism of GAT makes each detection head calculate the pairwise attention coefficients among all connected nodes. Considering the inherent high-dimensional structure in HSIs, this huge computational burden is obviously not conducive to engineering applications. Hence, inspired by Srinivasa et al. [45], we propose an SGAT module that prunes graphs by spectral sparsification under the condition of guaranteeing graph connectivity. We construct a random sampling function to filter edges and generate a random subgraph. Specifically, the random subgraph is required not only to remove redundant edges but also to preserve enough spectrum information of the original graph without causing excessive loss in the process of updating node features. The feature update is carried out by calculating the attention coefficient of the remaining edges of an attention head. Finally, the features in h attention heads are averaged to get the final feature update matrix.

The sparsification operation of the original graph is based on the premise of spectral domain graph convolution. For the lth graph $\boldsymbol{\mathcal{G}}^{(l)}$ with N nodes, we assume that the convolution kernel is g_θ, and the product of feature input $\boldsymbol{x}$ in the spectral domain can be written as

$$g_\theta * \boldsymbol{x} = \boldsymbol{U} g_\theta(\lambda)\boldsymbol{U}^{\mathrm{T}}\boldsymbol{x} \tag{5}$$

where $\boldsymbol{U}$ is the eigenvector matrix and λ is the diagonal matrix that is composed of the eigenvalues of the normalized Laplacian matrix.

In order to keep the spectral information of the original graph as much as possible, that is, the spectral content between the Laplacian matrix $\boldsymbol{L}_S$ of the sparsified subgraph and $\boldsymbol{L}_O$ of the original graph is similar, we refer to the ideas of [46] to set a correlation intensity variable $R(v_i, v_j)$ as the criterion for selecting edges. Mathematically, it is defined as

$$R(v_i, v_j) = \big(\boldsymbol{x}_{v_i} - \boldsymbol{x}_{v_j}\big)^{\mathrm{T}}\boldsymbol{L}^{\dagger}\big(\boldsymbol{x}_{v_i} - \boldsymbol{x}_{v_j}\big) \tag{6}$$

where v_i and v_j are the nodes position, and $\boldsymbol{x}_{v_i}$ and $\boldsymbol{x}_{v_j}$ are the standard basis vectors with 1. $\boldsymbol{L}^{\dagger}$ is the pseudoinverse of the graph Laplacian matrix [46], which is explained in [46] to approximate the spectral content by a distribution proportional to the correlation intensity associated with the edge. That is,

$$(1-\varepsilon)\lambda_i(\boldsymbol{L}_O) \le \lambda_i(\boldsymbol{L}_S) \le (1+\varepsilon)\lambda_i(\boldsymbol{L}_O) \tag{7}$$

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Algorithm 2 Edge Random Sampling

Input: Original graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, Initial sparse graph of $S(\mathcal{V}' = \mathcal{V}, \mathcal{E}' = NULL)$, set the number of edges to sample to $q = \max(M, \text{int}(0.5N \log N/\varepsilon^2)$.

```
b ∈ E, 0 < ε < 1 for each edge
P_b represents the probability that this edge will remain.
We use the algorithm in [46] to calculate the strength of the association for each edge b.
for i ∈ E do
    An edge b is randomly selected from the original graph, and the probability P_b of the selected edge is proportional to the product of the association strength and the weight value
    if selected edge b ∉ E' then
        Add edge b to E', set the weight of this edge to Z_b/qP_b
    else
        Add Z_b/qP_b weights to existing edge weights
    end if
end for
return S(V', E')
```

Output: $S(\mathcal{V}', \mathcal{E}')$

where ε is a pruning parameter whose value determines the tightness of the edge selection constraint, and $\lambda_i(\cdot)$ means taking the ith eigenvalue of some matrix.

The edge random sampling process generated by the correlation intensity is described in pseudocode of Algorithm 2. Concretely, we select any two nodes in the lth graph $\mathcal{G}^{(l)}$ with N nodes randomly and calculate intensity R of the selected nodes. $\boldsymbol{Z}_b/q\boldsymbol{P}_b$ is used as the weight to decide whether to add the selected edge to the sparse subgraph $\mathcal{S}^{(l)}$. Here, we define $\boldsymbol{Z}_b$ as the original weight of the selected edge and $\boldsymbol{P}_b$ as the selecting probability, which is proportional to the product of $\boldsymbol{Z}_b R$. If an edge is repeatedly selected, the weight of the original edge is self-added; inversely, the edge is created in $\mathcal{S}^{(l)}$ if a new edge is selected. Then, we take $q = O(N \log N/\varepsilon^2)$ as the edge selecting times, and the above steps are carried out independently for q times until q edges are selected. Owing to the selection of ε that determines the sparsified approximation between $\mathcal{G}^{(l)}$ and $\mathcal{S}^{(l)}$, we manually define the required ε to generate different numbers of edges, so as to obtain corresponding computational complexity. The complexity of computing all edges to form a sparse subgraph $\mathcal{S}^{(l)}$ is $O(M \log N)$ with M edges.

To apply the above $\mathcal{S}^{(l)}$ to a GAT, the random walk normalized Laplacian matrix $\boldsymbol{L}_R$ is adopted to redefine the spectral domain convolution operation

$$\hat{g}_\theta * \boldsymbol{x} = \boldsymbol{D}^{-1/2}\boldsymbol{U}\hat{g}_\theta(\lambda_R)\boldsymbol{U}^{\mathrm{T}}\boldsymbol{D}^{-1/2}\boldsymbol{x} \tag{8}$$

where λ_R is a diagonal matrix that is composed of the eigenvalues of $\boldsymbol{L}_R$ and $\boldsymbol{D}$ is a degree matrix.

The transformation between $\boldsymbol{L}_R$ and the symmetric normalized graph Laplacian matrix $\boldsymbol{L}_N$ can be expressed as

$$\boldsymbol{L}_R = \boldsymbol{I} - \boldsymbol{D}^{-1}\boldsymbol{A} = \boldsymbol{D}^{-1/2}\boldsymbol{L}_N\boldsymbol{D}^{1/2} \tag{9}$$

where $\boldsymbol{I}$ is the identity matrix.

Each layer in a GAT defines a new layer-dependent graph adjacency matrix $\boldsymbol{A}^{(l)}$ and the corresponding degree matrix $\boldsymbol{D}^{(l)}$. Then, $\boldsymbol{A}^{(l)}$ and $\boldsymbol{D}^{(l)}$ are used to calculate the first-order approximation of the convolution defined in (8) for each layer, so as to obtain the features $\boldsymbol{H}^{(l+1)}$ of the next layer

$$\boldsymbol{H}^{(l+1)} = \delta((\boldsymbol{D}^{(l)})^{-1}\boldsymbol{A}^{(l)}\boldsymbol{H}^{(l)}\boldsymbol{W}^{(l)}) \approx \delta(\hat{g}_\theta * \boldsymbol{H}^{(l)}) \tag{10}$$

where $\boldsymbol{H}^{(l)}$ is the features of the current layer and $\hat{g}_\theta$ is the convolution kernel characterized by $(\boldsymbol{D}^{(l)})^{-1}\boldsymbol{A}^{(l)}$. $\delta(\cdot)$ represents the activation function, which is assumed to be continuous with the Lipschitz constant.

Let $\boldsymbol{L}_{\mathrm{OS}}$ and $\boldsymbol{L}_{\mathrm{SS}}$ denote the symmetric normalized Laplacian matrixes of $\mathcal{G}^{(l)}$ and $\mathcal{S}^{(l)}$; $\boldsymbol{L}_{\mathrm{OR}}$ and $\boldsymbol{L}_{\mathrm{SR}}$ represent the random walk normalized Laplacian matrixes of the above graphs, separately. Then, the calculation features on $\mathcal{G}^{(l)}$ are derived as

$$\begin{aligned}\boldsymbol{H}^{(l)} &= \alpha((\boldsymbol{L}_{\mathrm{OR}} - \boldsymbol{I})\boldsymbol{H}^{(l)}\boldsymbol{W}^{(l)}) \\ &= \alpha((\boldsymbol{D}^{(l)})^{-1/2}(\boldsymbol{L}_{\mathrm{OS}} - \boldsymbol{I})(\boldsymbol{D}^{(l)})^{1/2}\boldsymbol{H}^{(l)}\boldsymbol{W}^{(l)})\end{aligned} \tag{11}$$

where $\alpha(\cdot)$ represents the activation function.

Furthermore, the features of spectral $\boldsymbol{L}_{\mathrm{SR}}$ calculated by a GAT model are given as follows:

$$\begin{aligned}\boldsymbol{H}_S^{(l)} &= \alpha\left((\boldsymbol{L}_{\mathrm{SR}} - \boldsymbol{I})\boldsymbol{H}^{(l)}\boldsymbol{W}^{(l)}\right) \\ &= \alpha\left(\left(\boldsymbol{D}_S^{(l)}\right)^{-1/2}(\boldsymbol{L}_{\mathrm{SS}} - \boldsymbol{I})\left(\boldsymbol{D}_S^{(l)}\right)^{1/2}\boldsymbol{H}^{(l)}\boldsymbol{W}^{(l)}\right).\end{aligned} \tag{12}$$

The error $\|\boldsymbol{L}_{\mathrm{OR}} - \boldsymbol{L}_{\mathrm{SR}}\|$ is limited and used to restrain the feature error $\boldsymbol{H}^{(l)} - \boldsymbol{H}_S^{(l)}$ between $\mathcal{G}^{(l)}$ and $\mathcal{S}^{(l)}$. The error $\|\boldsymbol{L}_{\mathrm{OR}} - \boldsymbol{L}_{\mathrm{SR}}\|$ calculated by a GAT model is formulized as

$$\begin{aligned}&\|\boldsymbol{L}_{\mathrm{OR}} - \boldsymbol{L}_{\mathrm{SR}}\| \\ &= \left\|(\boldsymbol{D}^{(l)})^{-1/2}\boldsymbol{L}_{\mathrm{OS}}(\boldsymbol{D}^{(l)})^{1/2} - \left(\boldsymbol{D}_S^{(l)}\right)^{-1/2}\boldsymbol{L}_{\mathrm{SS}}\left(\boldsymbol{D}_S^{(l)}\right)^{1/2}\right\| \\ &\le \|\boldsymbol{L}_{\mathrm{OS}} - \boldsymbol{L}_{\mathrm{SS}}\| \\ &\quad + \left\|\boldsymbol{L}_{\mathrm{SS}} - (\boldsymbol{D}^{(l)})^{1/2}\left(\boldsymbol{D}_S^{(l)}\right)^{-1/2}\boldsymbol{L}_{\mathrm{SS}}\left(\boldsymbol{D}_S^{(l)}\right)^{1/2}(\boldsymbol{D}^{(l)})^{-1/2}\right\|.\end{aligned} \tag{13}$$

Since $\boldsymbol{L}_{\mathrm{OS}}$ and $\boldsymbol{L}_{\mathrm{SS}}$ are both symmetric and positive semidefinite, we have

$$\begin{aligned}&\|\boldsymbol{L}_{\mathrm{OS}} - \boldsymbol{L}_{\mathrm{SS}}\| \\ &\quad = \max\left(\left|\lambda_{\max}(\boldsymbol{L}_{\mathrm{OS}} - \boldsymbol{L}_{\mathrm{SS}})\right|, \left|\lambda_{\min}(\boldsymbol{L}_{\mathrm{OS}} - \boldsymbol{L}_{\mathrm{SS}})\right|\right) \\ &\quad = \sup_{\boldsymbol{x}\in\mathbb{R}^N:\|\boldsymbol{x}\|=1}\left|\boldsymbol{x}^{\mathrm{T}}(\boldsymbol{L}_{\mathrm{OS}} - \boldsymbol{L}_{\mathrm{SS}})\boldsymbol{x}\right|\end{aligned} \tag{14}$$

where $\sup|\cdot|$ means taking the minimum upper bound, and then

$$\begin{aligned}&\sup_{\boldsymbol{x}\in\mathbb{R}^N:\|\boldsymbol{x}\|=1}\left|\boldsymbol{x}^{\mathrm{T}}(\boldsymbol{L}_{\mathrm{OS}} - \boldsymbol{L}_{\mathrm{SS}})\boldsymbol{x}\right| \\ &\quad \le \varepsilon\|\boldsymbol{L}_{\mathrm{OS}}\| + \left\|(\boldsymbol{D}^{(l)})^{-1/2}\left(\boldsymbol{D}_S^{(l)}\right)^{-1/2}\right. \\ &\qquad\qquad \left.\times\ \boldsymbol{L}_{\mathrm{SS}}\left(\boldsymbol{D}_S^{(l)}\right)^{-1/2}(\boldsymbol{D}^{(l)})^{-1/2} - \boldsymbol{L}_{\mathrm{SS}}\right\| \\ &\quad \le \varepsilon\|\boldsymbol{L}_{\mathrm{OS}}\| + 3\varepsilon\|\boldsymbol{L}_{\mathrm{OS}}\| = 4\varepsilon\|\boldsymbol{L}_{\mathrm{OS}}\|.\end{aligned} \tag{15}$$

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Algorithm 3 SGAT Module

Input: The current layer graph is $\mathcal{G}(\mathcal{V}, \mathcal{E})$, the number of attention heads is h, the activation function is δ, the weight matrix is $\boldsymbol{W}$, the input is $\boldsymbol{H} \in \mathbb{R}^{N\times D}$. M represents the number of edges that the current layer needs to take, $RES(\mathcal{E}, \boldsymbol{A}, M)$ represents for random sampling function.

1: **for** $i \in$ h **do**
2: $\mathcal{G}_i'(\mathcal{V}', \mathcal{E}') = RES(\mathcal{E}, \boldsymbol{A}, M)$, where $\mathcal{G}_i'(\mathcal{V}', \mathcal{E}')$ represents the sampled image
3: Calculate the coefficient matrix of the sampled graph $\mathcal{G}_i'(\mathcal{V}', \mathcal{E}')$:
4: $o_{j,k} = \exp\left(\delta\left(a^T\left[\boldsymbol{W}h_j || \boldsymbol{W}h_k\right]\right)\right)$
5: $\alpha \in \mathbb{R}^{N\times N}, \alpha(j,k) = o_{j,k} \Big/ \sum_{s\in N_j} o_{i,s}$
6: The updated feature matrix $\boldsymbol{H_i} = \delta(\alpha \boldsymbol{H}\boldsymbol{W})$
7: **end for**
8: Final $\boldsymbol{H}'$ is the average of the feature matrices obtained by each attention head, that is $\boldsymbol{H}' = \sum_{i=1}^{h} \boldsymbol{H_i}$

Output: $\boldsymbol{H}'$

For a very small value of ε, the assumption of $3\varepsilon^2 + \varepsilon^3 < \varepsilon$ is true. Then, (13) can be further calculated to obtain the upper Laplacian error of $\mathcal{G}^{(l)}$ and $\mathcal{S}^{(l)}$

$$\begin{aligned}
&\|\boldsymbol{L}_{\mathrm{OR}} - \boldsymbol{L}_{\mathrm{SR}}\| \\
&\quad \le \|\boldsymbol{L}_{\mathrm{OS}} - \boldsymbol{L}_{\mathrm{SS}}\| + \|\boldsymbol{L}_{\mathrm{SS}}\| \\
&\qquad \times \left(\left\| \boldsymbol{I} - \left(\boldsymbol{D}^{(l)}\right)^{1/2} \left(\boldsymbol{D}_S^{(l)}\right)^{-1/2} \right\| \right. \\
&\qquad\quad + \left\| \left(\boldsymbol{D}^{(l)}\right)^{1/2} \left(\boldsymbol{D}_S^{(l)}\right)^{-1/2} \right\| \\
&\qquad\quad \left. \times \left\| \boldsymbol{I} - \left(\boldsymbol{D}_S^{(l)}\right)^{1/2} \left(\boldsymbol{D}^{(l)}\right)^{-1/2} \right\| \right) \\
&\quad \le \|\boldsymbol{L}_{\mathrm{OS}} - \boldsymbol{L}_{\mathrm{SS}}\| + \varepsilon(1+\varepsilon)\|\boldsymbol{L}_{\mathrm{OS}}\| \\
&\quad = 4\varepsilon\|\boldsymbol{L}_{\mathrm{OS}}\| + \left(\varepsilon + \varepsilon^2\right)\|\boldsymbol{L}_{\mathrm{OS}}\| \\
&\quad \le 6\varepsilon\|\boldsymbol{L}_{\mathrm{OS}}\|. \qquad (16)
\end{aligned}$$

Finally, we can express the feature error $\boldsymbol{H}^{(l)} - \boldsymbol{H}_S^{(l)}$ as

$$\boldsymbol{H}^{(l)} - \boldsymbol{H}_S^{(l)} = 1_\delta 6\varepsilon \left\| (\boldsymbol{L}_{\mathrm{OS}} - \boldsymbol{L}_{\mathrm{SS}}) \boldsymbol{H}^{(l)} \boldsymbol{W}^{(l)} \right\| \qquad (17)$$

where 1_δ is the Lipschitz constant of $\delta(\cdot)$ and its value is made as 1.

The GATs from the original graph and the sparse subgraph have the same initial $\boldsymbol{W}^{(l)}$, and then, the corresponding feature updates are close in the sense of Frobenius norm. The spectrum sparse strategy proposed in this article can effectively retain the computational features required for updating the model. Algorithm 3 describes our SGAT module in pseudocode. For each head attention in the SGAT, the random sampling function makes the edges sparse and only finds the attention coefficients of the remaining edges to update the features. The average value of the features in the attention heads is calculated to get the final feature update matrix.

C. SIA Module

Relying on spatial information unduly cannot fully exploit the land cover information represented in HSIs. Considering taking full advantage of the correlation of HSI data bands, we design an SIA module to characterize the global relationship among spectral bands by constructing spectral graphs and obtain more discriminative spectral features for classification tasks. The nodes updated using the SIA contain more abundant global context information and then are provided to the next stage for learning the better specific expression.

To represent the similarity of each channel dimension at the lth layer, we establish the feature layer interdependence matrix $\boldsymbol{A}_F^{(l)}$, which can be written as

$$\boldsymbol{A}_F^{(l)} = \left(\boldsymbol{X}^{(l)}\right)^{\mathrm{T}} \times \boldsymbol{X}^{(l)} \qquad (18)$$

where $\boldsymbol{X}^{(l)} \in \mathbb{R}^{N\times D}$ denotes the original input features at the lth layer, N is the number of nodes, and D is the feature dimension of each node.

We use the identity matrix $\boldsymbol{I}$ to eliminate the correlation between nodes and the interdependence matrix itself. Then, $\boldsymbol{A}_F^{(l)}$ of the current layer can be further expressed as

$$\hat{\boldsymbol{A}}_F^{(l)} = \left(\boldsymbol{I} - \boldsymbol{A}_F^{(l)}\right). \qquad (19)$$

When the nodes are updated by the SIA module, the convolution operation on the original features can be realized to obtain the new features $\boldsymbol{M}_F^{(l)}$, i.e.,

$$\boldsymbol{M}_F^{(l)} = \boldsymbol{X}^{(l)} \hat{\boldsymbol{A}}_F^{(l)} \boldsymbol{W}_F^{(l)} \qquad (20)$$

where $\boldsymbol{W}_F^{(l)} \in \mathbb{R}^{D\times D}$ represents the trainable weights specific to this layer, and $\boldsymbol{W}_F^{(l)}$ is randomly initialized and optimized by gradient descent during end-to-end training.

The SIA can further effectively enhance the representation of the original spectra by constructing spectral graphs to characterize the global relationship among spectral bands. The feature $\boldsymbol{M}_F^{(l)}$ enhanced by SIA is consistent with the original input $\boldsymbol{X}^{(l)}$ at the current layer in terms of size and dimension, so it is compatible with SGAT.

In architectural composition, we use the SIA in parallel with the SGAT to form an SSCU, so as to complement the distribution features that are extracted from the two branches, as shown in Fig. 4. We assign different weights to the two branches for different degrees of scaling, so as to better perform feature fusion and integrate the model, which is represented as follows:

$$\boldsymbol{X}^{(l+1)} = (1-\alpha)\mathrm{SGAT}\left(\boldsymbol{X}^{(l)}\right) + \alpha \mathrm{SIA}\left(\boldsymbol{X}^{(l)}\right) \qquad (21)$$

where $\alpha \in (0, 1)$ represents the proportion allocated.

D. PLT Module

The features acquired in the last stage of the decoder are consistent with the same dimension as those obtained from the initial HSI squeezed, which is represented as superpixels. Generally, the superpixel segmentation method with certain errors and rough edge features cannot generate finer individual features for each pixel, which limits the further improvement of classification accuracy. Therefore, the PLT module aims to

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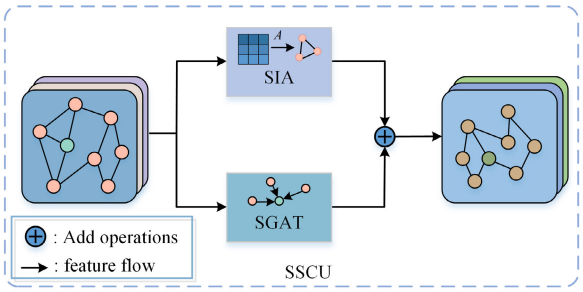


Fig. 4. Node features updating illustration of SSCU with the original input (left graph nodes) and new features (right graph nodes).

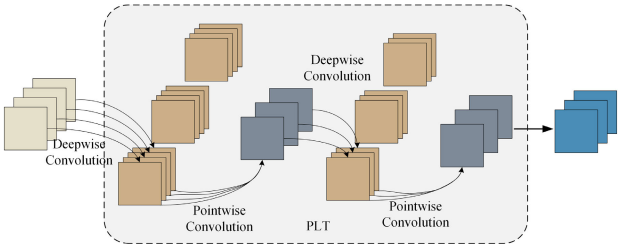


Fig. 5. Dashed box indicates the proposed PLT, the scattered arrows indicate the depthwise convolution operation, and the merged arrows represent the pointwise convolution operation.

perform pixel-level feature extraction for the final multiscale outputs to improve the refinement and generalization ability of our network. The features extracted from the encoder–decoder PLT can be inputted into various classifiers for classification, such as EPF developed by Kang et al. [47]. In this study, to preserve the integrity and high-dimensional representation of the features, we utilize fully connected layers to classify the features extracted from PLT. The structure of PLT is exhibited in Fig. 5, which consists of two layers of 3×3 depth-separable convolution to reduce redundancy among spatial and spectral dimensions and improve the spatial resolution of features in the process of graph convolution.

E. Loss Function

The cross-entropy loss function is employed to train the model, and the weight and deviation of our model are updated by backpropagation and stochastic gradient descent. The formula is written as

$$\mathcal{L}_{\text{loss}} = -\frac{1}{N_s}\sum_{1}^{N_s}\sum_{1}^{C_s} L_c^t \log\left(\hat{L}_c^t\right) \tag{22}$$

where L_c^t and $\hat{L}_c^t$ represent the predicted and true pixel values belonging to class c, separately. N_s is the total number of samples, and C_s represents the number of categories.

IV. Experimental Analysis

In this section, our proposed MSG-SGAT is compared with some state-of-the-art methods on four real public hyperspectral datasets. Several widely used metrics, such as precision, overall accuracy (OA), average accuracy (AA), and kappa statistics (KPP), are adopted in this work. Network depth experiments conducted to measure the classification performance are followed by comparison and ablation experiments to validate the efficacy of our proposed modules. Then, the impact analysis of combining the SGAT and PLT modules with different ratios on the performance improvement is performed. Next, the learning ability of our proposed model for a small number of samples and for the graphs on these four datasets is tested. In addition, we conduct the comparison experiment with the proposed SGAT and traditional GAT to verify its effectiveness, and finally, the influence of the initial graph on network performance and computing time under different initial superpixel conditions is analyzed.

TABLE I
Parameters of the Datasets Used in This Article

Dataset	Total	Size	Class	Band	Train	Val	Test
IN	10249	145×145	16	200	5%	1%	94%
PU	41447	610×340	9	103	1%	1%	98%
SV	54129	512×217	16	204	1%	1%	98%
HU	15029	349×1905	15	144	5%	1%	94%

We implement all the models on the open-source PyTorch 1.2.0 framework executed on a 64-bit Ubuntu computer with 24-GB GeForce P6000 GPU and Intel Core i9-11900K CPU with 64-GB memory. The comparison models are implemented in their original environments without any additions, and all experiments are conducted in the same environment. Furthermore, the Adam function is chosen as the optimizer, the learning rate is set as 0.0001, and the epoch is 700. Since graph convolution operations usually achieve good performance without deep structure, the multistage encoder and decoder are designed as four stages in this article. Concretely, we set 1024 nodes (i.e., superpixels) at the first stage and halve them with the stage increasing, subsequently. Meanwhile, ε of the SGAT is assigned as 0.5. It is important to emphasize that all reported results are calculated from the average of six independent experiments in order to keep the quantitative data more representative.

The performance of all the HSIs classification methods is evaluated on the following four real public datasets in this article, namely, Indian Pines (IN), University of Pavia (PU), Salinas (SV), and Houston 2013 (HU) datasets. Specifically, all the parameters of labeled samples assigned to training, validation, and testing for each dataset are listed in Table I.

A. Comparison Experiments

Comparison experiments are conducted with nine existing advanced methods to verify the effectiveness of our proposed MSG-SGAT, including IRTS-NKLCMV [48], HybridSN [49], SPRN [50], FDGC [51], CEGCN [52], MSSGU [53], SAGE-A [54], WFCG [55], and SPGAT [38]. The experiment is repeated six times to report the mean and standard deviation of each indicator. Tables II–V list the quantitative accuracies of different methods on each dataset, in which the bold denotes the best results and the second-best is underlined. Moreover, the corresponding visual results are illustrated in Figs. 6–9.

As shown in Table II, our proposed method achieves the best classification performance among all methods in terms of OA, AA, and KPP on the IN dataset. The accuracies obtained by FDGC and HybridSN are relatively low, and we suspect that these CNN-based methods cannot be fully trained owing to their deep structure in the case of finite training samples. In comparison, the GNN-based methods, such as WFCG,

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TABLE II
CLASSIFICATION RESULTS OF VARIOUS METHODS ON THE IN DATASET WITH 5% TRAINING SAMPLES

	Mixed Pixel Method	CNN Methods			GCN Methods		GAT Methods			
Class	**IRTS-NKLCMV**	**HybridSN**	**SPRN**	**FDGC**	**CEGCN**	**MSSGU**	**SSGE-A**	**WFCG**	**SPGAT**	**Ours**
1	95.72±1.56	65.00±7.96	89.80±1.44	70.11±18.83	74.01±16.23	88.00±4.14	71.04±8.69	**100.0±0**	62.92±5.67	91.20±4.14
2	92.20±1.57	92.87±1.13	97.58±0.66	91.12±8.67	96.73±1.38	**98.96±0.57**	89.76±2.49	95.73±0.67	87.68±2.50	98.94±0.57
3	95.27±2.36	94.90±3.31	89.67±0.79	90.87±5.60	96.91±1.39	98.34±0.44	88.23±2.72	**98.79±0.96**	90.13±5.53	98.52±0.44
4	98.72±0.54	84.53±8.95	79.54±43.90	87.34±7.71	89.61±12.16	**100.0±0**	82.21±3.97	99.10±0.95	84.34±1.95	**100.0±0**
5	95.71±1.51	98.34±1.59	91.81±2.02	90.84±8.41	95.79±3.60	**99.57±0.32**	88.38±2.38	93.42±1.53	91.04±1.34	99.52±0.38
6	96.65±1.82	98.21±1.76	99.31±0.27	94.63±4.08	99.49±0.51	99.86±0.12	98.35±0.21	**99.91±0.08**	98.03±0.93	99.68±0.27
7	95.52±9.01	80.00±9.88	**100.0±0**	38.33±20.33	69.11±30.84	96.52±1.94	99.13±0.52	**100.0±0**	92.25±2.02	95.65±3.66
8	97.90±1.67	99.95±0.09	**100.0±0**	98.01±1.75	**100.0±0**	**100.0±0**	99.20±0.13	**100.0±0**	99.80±0.24	**100.0±0**
9	94.10±8.47	**98.95±2.35**	35.55±9.87	26.31±23.40	30.53±19.48	66.67±18.00	74.44±9.98	95.55±2.48	75.18±3.72	96.66±0.05
10	92.64±2.71	96.77±1.17	97.33±1.00	92.18±4.65	95.37±3.48	97.36±0.68	94.50±2.91	91.98±0.97	91.75±1.34	**98.46±0.68**
11	94.17±1.25	97.55±0.72	95.60±1.42	94.79±2.90	98.08±0.91	98.79±0.37	94.58±2.24	96.95±1.58	88.45±2.94	**99.94±0.37**
12	94.92±2.67	87.49±2.36	97.66±1.63	81.35±11.01	97.44±1.78	96.35±1.14	92.47±5.76	97.13±0.55	86.43±2.82	**98.05±1.14**
13	95.30±3.12	92.82±2.12	99.64±0.81	96.12±6.61	99.80±0.27	**100.0±0**	96.45±2.49	99.79±0.28	99.24±0.14	**100.0±0**
14	99.07±0.39	99.31±0.70	98.73±0.50	98.27±1.49	99.54±0.19	**100.0±0**	98.59±0.80	99.75±0.17	97.35±0.05	99.74±0.43
15	99.03±0.62	87.30±4.85	99.03±1.00	85.25±7.54	95.59±4.84	**100.0±0**	92.98±0.91	99.06±0.60	90.25±2.45	99.72±0.13
16	89.34±3.10	82.04±7.93	**99.76±0.53**	83.73±10.40	95.70±3.16	89.89±3.27	81.80±2.20	99.54±0.63	95.79±0.87	98.38±1.23
OA(%)	95.18±0.71	95.33±0.37	96.47±0.26	92.32±4.15	97.18±0.52	98.76±0.23	93.28±0.73	97.23±0.47	91.14±1.89	**98.98±0.14**
AA(%)	97.54±0.59	94.89±0.52	93.18±1.42	82.67±8.68	89.61±2.35	96.14±0.97	87.04±0.98	97.92±0.25	88.37±1.04	**98.43±0.24**
KPP(×100)	94.53±0.80	94.66±0.42	95.99±0.30	91.24±4.74	96.78±0.60	98.59±0.16	92.36±0.73	96.84±0.53	89.91±1.54	**98.92±0.18**

TABLE III
CLASSIFICATION RESULTS OF VARIOUS METHODS ON THE PU DATASET WITH 1% TRAINING SAMPLES

	Mixed Pixel Method	CNN Methods			GCN Methods		GAT Methods			
Class	**IRTS-NKLCMV**	**HybridSN**	**SPRN**	**FDGC**	**CEGCN**	**MSSGU**	**SAGE-A**	**WFCG**	**SPGAT**	**Ours**
1	97.05±0.48	96.03±1.89	99.49±0.07	97.10±1.88	99.50±0.44	97.91±0.28	96.98±2.57	98.92±0.41	96.07±0.37	**99.66±0.52**
2	99.56±0.19	99.68±0.23	99.89±0.04	99.83±0.13	**99.99±0.01**	99.98±0.01	98.54±1.69	99.97±0.02	97.68±0.23	99.97±0.01
3	93.03±2.31	83.13±5.42	97.30±0.32	87.33±6.12	99.05±1.07	**100.0±0**	99.02±0.76	**100.0±0**	94.93±0.24	**100.0±0**
4	94.61±1.25	88.22±3.39	98.47±0.28	90.01±3.75	96.19±1.87	97.65±0.86	97.99±1.96	**98.50±0.22**	91.50±1.55	97.85±0.51
5	99.65±0.26	98.03±2.28	99.89±0.03	98.54±2.57	**100.0±0**	**100.0±0**	99.84±0.09	**100.0±0**	99.73±0.09	**100.0±0**
6	99.26±0.41	98.15±2.47	99.74±0.27	99.88±0.12	99.98±0.04	99.99±0.02	99.28±0.43	**100.0±0**	97.63±1.04	**100.0±0**
7	99.42±0.63	94.96±4.68	99.89±0.11	98.98±0.73	99.95±0.10	97.28±5.57	99.76±0.07	99.91±0.10	99.46±0.50	**100.0±0**
8	91.29±0.91	91.50±2.81	97.72±0.88	95.78±1.95	99.33±0.67	**100.0±0**	94.41±2.21	98.91±0.91	97.45±1.33	99.97±0.03
9	96.61±1.29	69.82±5.02	99.24±0.34	79.59±7.54	99.72±0.19	99.31±0.33	97.70±1.50	99.65±0.38	90.51±0.49	**99.76±0.20**
OA(%)	97.70±0.15	95.74±0.41	99.39±0.08	97.23±0.70	99.53±0.13	99.39±0.08	98.20±1.01	99.60±0.05	97.30±0.22	**99.77±0.07**
AA(%)	96.96±0.38	93.67±0.93	99.07±0.12	94.11±0.76	99.30±0.26	99.12±0.59	97.57±1.07	99.53±0.09	96.11±0.28	**99.59±0.27**
KPP(×100)	96.94±0.20	94.33±0.54	99.18±0.11	96.32±0.93	99.38±0.17	99.01±0.64	98.15±0.41	99.48±0.07	96.05±0.29	**99.70±0.10**

TABLE IV
CLASSIFICATION RESULTS OF VARIOUS METHODS ON THE SV DATASET WITH 1% TRAINING SAMPLES

	Mixed Pixel Method	CNN Methods			GCN Methods		GAT Methods			
Class	**IRTS-NKLCMV**	**HybridSN**	**SPRN**	**FDGC**	**CEGCN**	**MSSGU**	**SAGE-A**	**WFCG**	**SPGAT**	**Ours**
1	99.99±0.01	**100.0±0**	**100.0±0**	99.83±0.17	99.93±0.16	**100.0±0**	99.37±0.23	**100.0±0**	99.62±0.28	**100.0±0**
2	99.99±0.02	**100.0±0**	**100.0±0**	99.98±0.03	**100.0±0**	99.99±0.22	99.55±0.28	**100.0±0**	99.95±0.03	99.99±0.01
3	99.94±0.06	99.93±0.11	**100.0±0**	99.92±0.16	**100.0±0**	**100.0±0**	99.58±0.42	99.89±0.23	99.86±0.21	**100.0±0**
4	98.08±1.43	95.27±3.48	99.40±0.12	97.85±2.35	99.38±0.52	99.34±0.21	97.79±2.16	**99.60±0.19**	99.28±0.45	99.56±0.38
5	99.36±0.37	98.75±0.86	**99.86±0.09**	98.48±1.93	98.92±0.86	98.97±0.16	98.89±0.77	98.98±0.31	97.91±1.77	98.93±0.01
6	99.97±0.02	99.91±0.16	**100.0±0**	99.88±0.23	**100.0±0**	**100.0±0**	99.49±0.67	99.96±0.08	99.87±0.30	**100.0±0**
7	99.77±0.31	99.99±0.01	**100.0±0**	99.84±0.12	99.99±0.03	**100.0±0**	98.09±1.67	**100.0±0**	99.61±0.08	99.97±0.02
8	96.07±1.27	98.67±0.98	97.46±0.57	**99.27±0.75**	99.02±0.64	98.79±0.15	96.39±2.49	99.00±0.72	92.26±4.57	98.93±0
9	99.94±0.08	**100.0±0**	**100.0±0**	99.99±0.01	**100.0±0**	**100.0±0**	95.14±3.61	**100.0±0**	99.88±0.12	**100.0±0**
10	99.25±0.77	99.34±0.39	98.83±0.44	98.98±1.08	98.30±1.78	**99.99±0.01**	98.44±1.29	97.83±1.01	96.66±1.78	99.98±0.01
11	99.93±0.06	94.49±2.47	**99.98±0.04**	99.52±0.28	99.96±0.08	99.94±0.13	99.53±0.55	99.79±0.15	98.93±0.77	99.88±0.01
12	99.77±0.08	99.39±1.33	99.98±0.05	99.74±0.32	**100.0±0**	99.89±0.06	98.08±0.84	**100.0±0**	99.30±0.59	**100.0±0**
13	99.52±0.31	99.23±0.80	99.59±0.29	94.26±7.32	**99.95±0.06**	95.65±1.26	98.32±1.11	99.53±0.34	92.63±3.64	98.95±0.11
14	99.34±0.40	**99.83±0.19**	99.61±0.23	98.16±0.92	98.34±2.12	97.42±0.28	98.95±0.32	98.34±1.37	98.17±0.87	97.29±0.37
15	94.40±2.07	97.05±1.16	98.04±0.24	99.73±0.19	99.22±0.37	**100.0±0**	97.05±2.57	99.37±0.47	94.77±2.01	99.99±0
16	99.73±0.35	99.86±0.26	99.81±0.19	98.69±1.12	99.32±0.88	**100.0±0**	98.24±0.92	99.96±0.07	99.32±0.13	99.47±0.02
OA(%)	98.22±0.57	98.94±0.31	99.09±0.11	99.39±0.24	99.46±0.16	99.55±0.10	98.25±0.57	99.46±0.09	97.17±0.54	**99.61±0.07**
AA(%)	99.24±0.26	99.07±0.09	99.53±0.04	99.01±0.48	99.52±0.14	99.37±0.10	98.26±0.30	99.51±0.10	98.41±0.22	**99.56±0.04**
KPP(×100)	98.01±0.63	98.82±0.34	98.99±0.12	99.32±0.27	99.40±0.18	99.50±0.06	98.05±0.56	99.40±0.11	96.85±0.60	**99.57±0.03**

CEGCN, and MSSGU, achieve preferable precision, which is mainly due to the preponderance of combining the pixel and superpixel features, and learning adequacy of the complementary spatial–spectral features. IRTS-NKLCMV also performs well in these comparison experiments and achieves satisfactory performance. In terms of OA, it is 2.86% and 4.04% higher than FDGC and SPGAT pertaining to the GNN-based methods, respectively. On the basis of combining interactive features, our MSG-SGAT adaptively merges features suitable for various land covers at each stage under the guidance of multilevel superpixels. The performance of this method in OA and KPP is 0.29% and 0.33% higher than that of MSSGU, respectively, and 0.54% higher than the suboptimal value in AA. Our proposed method outperforms the other baseline method, which is proven by all these statistics.

Our proposed method on a sample instance of the IN results in a smoother visual effect and fewer errors than other comparative methods visually, which is closer to the GT, as shown in Fig. 6. Furthermore, Grass-p, Grass-m, and Stone are chosen as the validation categories to verify the

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TABLE V
CLASSIFICATION RESULTS OF VARIOUS METHODS ON THE HU DATASET WITH 5% TRAINING SAMPLES

	Mixed Pixel Method	CNN Methods			GCN Methods		GAT Methods			
Class	**IRTS-NKLCMV**	**HybridSN**	**SPRN**	**FDGC**	**CEGCN**	**MSSGU**	**SAGE-A**	**WFCG**	**SPGAT**	**Ours**
1	98.11±0.09	96.72±1.02	97.40±0.78	97.03±1.22	98.00±0.11	98.02±0.04	97.14±0.91	95.71±1.42	97.70±0.44	**99.66±0.25**
2	97.78±0.65	97.75±0.32	96.26±0.29	98.23±0.68	98.36±0.08	96.54±0.61	98.21±1.14	96.61±2.67	96.28±0.46	**99.16±0.67**
3	**100±0**	99.37±0.52	98.89±0.29	99.45±0.42	99.95±0.10	**100±0**	99.50±0.24	**100±0**	98.83±0.39	**100±0**
4	99.2±0.49	96.05±1.53	96.44±0.53	97.18±1.62	99.51±0.18	98.56±0.58	96.55±0.84	96.47±2.61	96.29±0.86	**99.79±0.24**
5	98.79±0.85	98.94±0.70	99.78±0.21	99.88±0.14	99.48±0.87	99.39±0.87	**99.93±0.06**	99.89±0.25	99.84±0.13	99.46±0.48
6	99.88±0.25	98.14±1.07	96.51±0.80	98.15±1.00	99.90±0.15	**100±0**	96.65±3.42	98.46±2.00	96.76±0.39	96.44±1.72
7	97.81±1.09	88.78±2.04	94.24±0.89	96.15±0.59	96.07±0.64	96.80±1.21	95.99±0.75	**99.55±0.22**	93.75±0.69	97.97±0.88
8	96.35±1.23	95.90±1.29	97.68±1.11	**98.01±0.84**	96.17±1.20	97.55±1.51	97.19±1.81	94.62±1.42	97.16±0.58	97.08±0.43
9	95.55±0.91	90.80±2.40	**98.68±0.86**	95.92±1.95	97.46±0.74	97.86±0.93	95.06±2.01	98.14±1.11	92.76±0.55	97.10±0.55
10	99.39±0.83	96.41±1.65	**99.99±0.01**	99.02±0.27	99.01±0.19	98.69±1.00	98.37±1.17	98.78±0.45	99.98±0.04	99.74±0.32
11	99.73±0.59	96.39±1.47	99.03±0.63	99.40±0.23	**99.89±0.16**	99.86±0.28	98.95±0.65	99.28±0.44	99.25±0.81	99.63±0.43
12	97.79±0.47	96.90±0.95	95.71±0.97	97.90±1.36	95.89±0.37	97.77±0.65	**98.69±0.32**	95.79±0.91	95.72±0.39	98.38±0.57
13	96.63±1.04	89.47±3.92	85.29±1.69	94.97±1.99	91.73±1.02	96.23±0.52	95.58±3.45	**98.21±0.90**	84.80±0.67	96.40±0.95
14	**100±0**	99.49±0.55	99.92±0.16	99.17±1.78	**100±0**	**100±0**	98.93±1.73	**100±0**	**100±0**	**100±0**
15	**100±0**	99.54±0.39	98.44±0.49	98.28±2.12	99.92±0.09	99.98±0.03	99.14±0.54	**100±0**	97.63±1.20	99.71±0.39
OA(%)	98.27±0.28	95.82±0.65	96.80±0.15	97.91±0.47	98.07±0.19	98.31±0.17	97.71±0.41	97.82±0.54	96.72±0.18	**98.80±0.09**
AA(%)	98.41±0.26	95.83±0.68	96.12±0.09	97.97±0.33	97.85±0.19	98.35±0.16	97.70±0.52	97.96±0.48	96.05±0.10	98.44±0.23
KPP(×100)	98.13±0.31	95.48±0.70	96.54±0.16	97.74±0.51	97.91±0.20	97.00±2.84	97.53±0.44	97.64±0.59	96.45±0.19	**98.70±0.10**

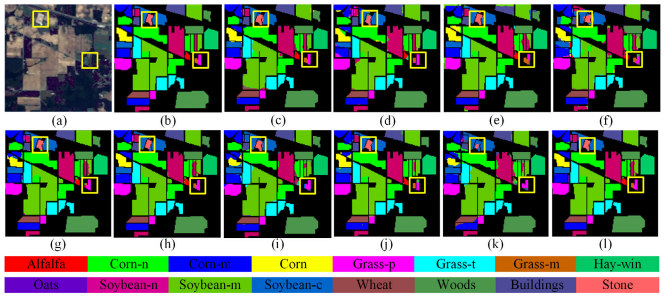


Fig. 6. Classification results of contrastive methods on the IN dataset. (a) False-color image. (b) GT. (c) IRTS-NKLCMV (OA = 95.18). (d) HybridSN (OA = 95.33). (e) SPRN (OA = 96.47). (f) FDGC (OA = 92.32). (g) CEGCN (OA = 97.18). (h) MSSGU (OA = 98.76). (i) SAGE-A (OA = 93.28). (j) WFCG (OA = 97.23). (k) SPGAT (OA = 91.14). (l) Ours (OA = 98.98).

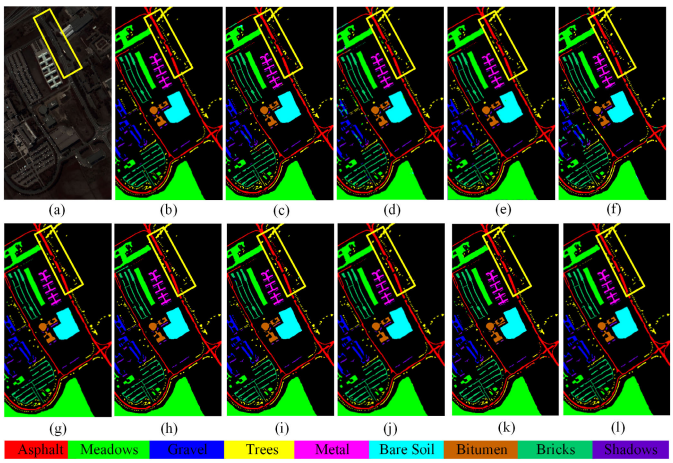


Fig. 7. Classification results of contrastive methods on the PU dataset. (a) False-color image. (b) GT. (c) IRTS-NKLCMV (OA = 97.70). (d) HybridSN (OA = 95.74). (e) SPRN(OA = 99.39). (f) FDGC (OA = 97.23). (g) CEGCN(OA = 99.53). (h) MSSGU(OA = 99.38). (i) SAGE-A (OA = 98.20). (j) WFCG (OA = 99.60). (k) SPGAT (OA = 97.30). (l) Ours (OA = 99.77).

intraclass and interclass differences identify the ability of the MSG-SGAT. Compared with other results illustrated in the highlighted areas of Fig. 6, our method produces fewer errors, and the category boundary is more consistent with the actual cover boundary. However, SAGE-A, FDGC, and SPGAT produce more classification errors because they are

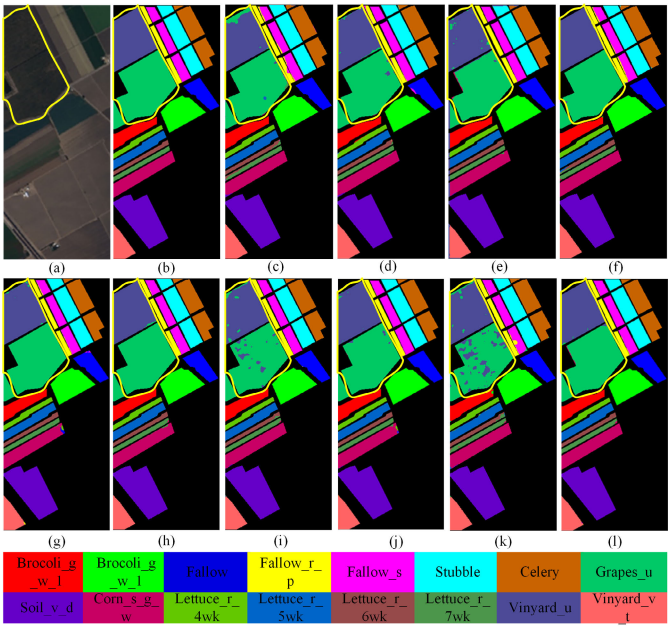


Fig. 8. Classification results of contrastive methods on the SV dataset. (a) False-color image. (b) GT. (c) IRTS-NKLCMV (OA = 98.22). (e) HybridSN (OA = 98.94). (g) SPRN (OA = 99.09). (j) FDGC (OA = 99.39). (h) CEGCN (OA = 99.46). (f) MSSGU (OA = 99.55). (d) SAGE-A (OA = 98.25). (i) WFCG (OA = 99.46). (k) SPGAT (OA = 97.17). (l) Ours (OA = 99.61).

not sufficiently trained to capture global context information effectively.

Similarly, Table III reports the classification performance on the PU dataset, and the performance of IRTS-NKLCMV is demonstrated to be still slightly higher than that of FDGC and SPGAT. Meanwhile, the method proposed in this article is also effective in various indicators even though most of the land covers in the PU samples are discontinuous and fragmented. The GNN-based methods, such as CEGCN and WFCG, can adaptively capture the features of irregular regions and obtain more stable classification results. However, the CNN-based methods, including FDGC and HybridSN, have relatively low accuracy because they are unable to extract distance-dependent spatial information by adopting fixed convolution kernels on regular grids. In contrast, our MSG-SGAT can effectively

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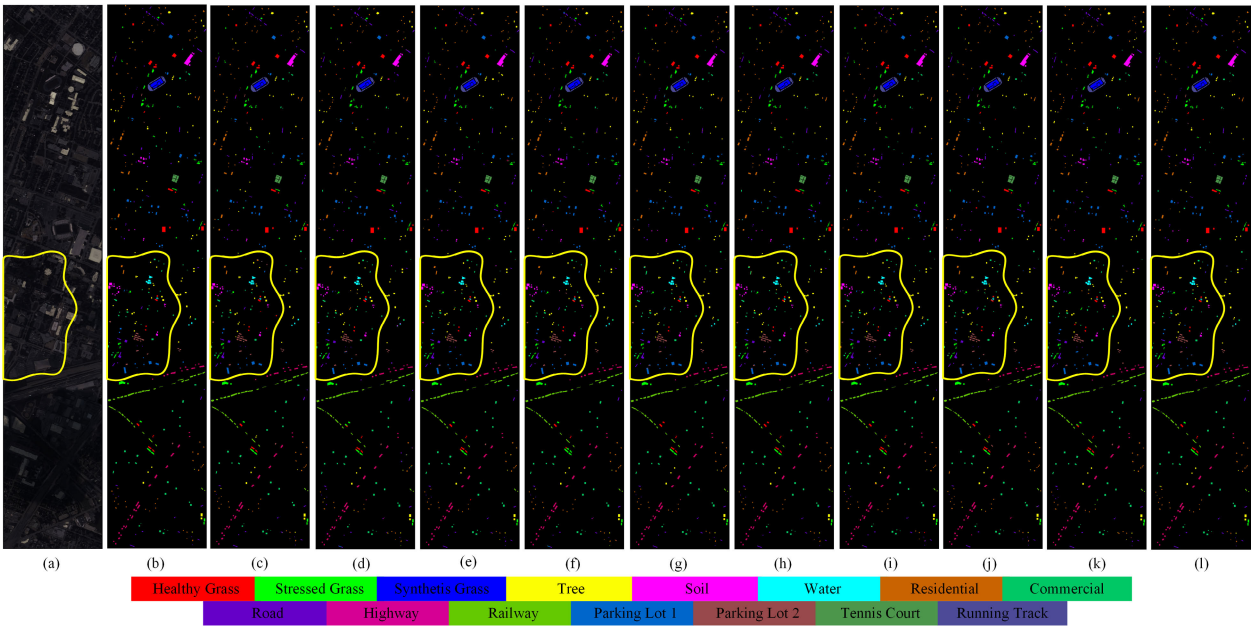


Fig. 9. Classification results of contrastive methods on the HU dataset. (a) False-color image. (b) GT. (c) IRTS-NKLCMV (OA = 98.27). (d) HybridSN (OA = 95.82). (e) SPRN (OA = 96.80). (f) FDGC (OA = 97.91). (g) CEGCN (OA = 98.07). (h) MSSGU (OA = 98.31). (i) SAGE-A (OA = 97.71). (j) WFCG (OA = 97.82). (k) SPGAT (OA = 96.72). (l) Ours (OA = 98.80).

sense the topology of irregular regions in sample instances and capture the context distribution around objects flexibly, so as to ensure the region homogeneity and the boundary continuity. Similarly, the visuals in Fig. 7 also demonstrate the standpoint mentioned above. Taking the highlighted area in Fig. 7 as examples, compared with other ground covers, the asphalt road (red lump) presents a narrow strip. Meanwhile, due to its proximity to the surrounding trees (yellow lump), it is prone to discontinuity and misclassification. The regional classification results of this category obtained by our method show that the boundary is smooth and clear. There are no significant misclassifications in the land covers corresponding to all categories, especially in narrow areas. In addition, SAGE-A, HybridSN, FDGC, and SPGAT, due to their fixed form kernels, have poor performance and larger errors than our method when the same type of land cover is widely distributed, as shown in Fig. 7(d), (f), (i), and (k).

Table IV and Fig. 8 show the classification performance on the SV dataset quantitatively and qualitatively, and the sample instances presented are mainly composed of large areas of land cover. Compared with others, the method presented in this article has advantages for the classification of large land cover regions with fewer false pixels. Although HybridSN, SPRN, FDGC, and other CNN-based methods obtain good classification performance by adopting advanced modules, such as multiscale convolution, residual connection, and attention mechanism, the feature-capturing ability is limited. The network overfitting may still be caused in the case of finite training samples and deep structure, which affects the network efficiency in turn. It is worth mentioning that IRTS-NKLCMV combining NKLCMV, spatial filter, and IRTS also obtains good classification effect on the SV dataset, and this nonneural network theory is worthy of recommendation for the HSI classification task, which can be verified from Fig. 8(c). Our MSG-SGAT captures multiscale spatial and spectral context information in a hierarchical manner to perform pixel classification, which is especially suitable for such SV samples with large ground object distribution. As can be seen from the grape planting fields in the highlighted areas, our proposed network has a good ability to aggregate inner class information and also performs a superior distinction among similar categories.

Ulteriorly, our proposed method achieves excellent results on the HU dataset, as demonstrated by Table V and Fig. 9. The HU dataset contains a high proportion of discontinuous and fragmented land covers, which poses a significant challenge to HSI classification accuracy. However, our method is designed with a multilevel graph structure, which can effectively model complex spatial topologies of land covers with various shapes and sizes, thereby preserving class boundaries and improving classification accuracy. The superiority in OA, AA, and KPP indicators proves the effectiveness of our proposed method. In addition, the visualization results of the highlighted areas in Fig. 9 show that our MSG-SGAT can accurately identify small land covers, resulting in a smoother visual classification output, which is highly effective in addressing the challenge of irregular land cover classification. Comparatively, HybridSN has poorer performance than other methods, mainly due to its difficulty in capturing larger contextual information during the convolution process. This results in the loss of useful local details and spatial structure that are essential for accurate classification, especially for scattered region distributions in the HU samples.

In order to further analyze the land cover categories recognition ability of these methods, we randomly select a category (such as Bitumen; see highlighted areas in Fig. 10) from the PU dataset and visualize the learned feature mapping

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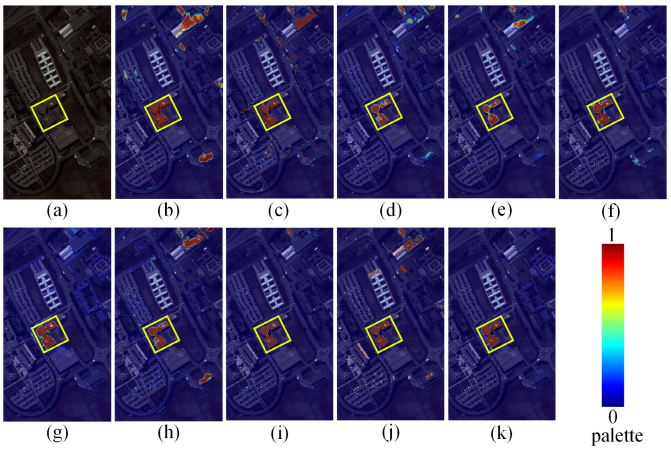


Fig. 10. Feature mapping visualization of the selected category in a PU sample. (a) PU (Bitumen). (b) IRTS-NKLCMV. (c) HybridSN. (d) SPRN. (e) FDGC. (f) CEGCN. (g) MSSGU. (h) SAGE-A. (i) WFCG. (j) SPGAT. (k) Ours.

corresponding to it. Due to noise pollution, other comparison methods will more or less lead to the misclassification of some small regions, blurred boundary details, and blurred boundary details, especially for the CNN-based methods, such as HybridSN. In contrast, the superpixel-guided GNN method reduces the misclassification of boundary regions by generating superpixel that reflects the general categories of actual land cover distribution. Ulteriorly, our proposed method preserves clearer regional edges by fusing multiscale spatial information with spectral context.

B. Stages' Influence

The influence of the stage number of the encoder and the decoder on the network classification performance is analyzed in this section. The graphs generated at different levels, as well as the spatial topology and scale available for fusion, are proportional to the stages of the U-shaped structure. However, for computational efficiency and optimal performance, the number of stages should be limited to a certain range. The stages are set as 1∼9 to test the performance of MSG-SGAT on the above four datasets, respectively, which are visualized in the form of a candle diagram in Fig. 11. The OA decreases with increasing stages, and the maximum values occur when the encoder and the decoder have four stages, which is because the relevant features from adjacent nodes are aggregated into the core node. Aggregating and transforming the feature information of neighbor nodes can promote classification performance in the shallow stage, but, in the deep stage, the high-order aggregated features aggravate the problem of oversmoothing and vanishing gradient. Moreover, the characteristics of different nodes tend to be homogenized with the deepening of network layers. Here, we set four stages in MSG-SGAT.

In addition, we observe that OA changes at a much higher rate in the IN and SV datasets than that in the PU. The samples of the IN and SV datasets are composed of larger scale land covers that are more sensitive to the aggregation of intraclass and interclass information, and the results are more susceptible to noise and lead to great changes in classification performance with stages changing. Meanwhile, the change rate of OA on

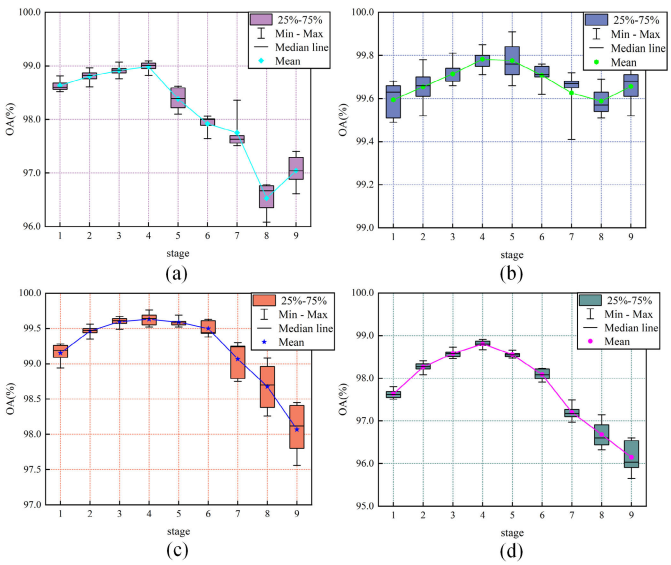


Fig. 11. Classification accuracy of our MSG-SGAT with different network stages on four public datasets. (a) IN. (b) PU. (c) SV. (d) HU.

the HU dataset also changed considerably with the deepening of layers owing to the scattered distribution of overlays.

C. Ablation Experiments

We conduct detailed ablation experiments for PLT, SIA modules, and skip connection to verify their effects on our model, which are described by OA on these four datasets in this article. Skip connection concatenates the encoder to the decoder with the same stage for information interaction. The connection ablation testing at each stage means that the skip connection of stage S is broken to prevent the features generated by this stage in the encoder for the feature fusion. The classification accuracies of different module combinations are recorded in Table VI, where ●/○ indicates that the module is adopted or not in the test, respectively.

After ablating the PLT, the OA on the four datasets decreases by 1.67%, 0.4%, 0.36%, and 1.44%, respectively, which indicates the importance of the PLT to the graph feature refinement. While adopting the SIA, the performance of our network is improved, which implies that the proposed SIA can enhance the spectral discrimination ability and optimize the network on the premise of retaining native features. Similarly, we observe that the combination of the SIA and PLT has a complementary effect to improve classification performance. In addition, skip connection also has a great effect on the classification. For instance, the first skip disconnection results in an evident performance decline, which shows that the features learned on low-level graphs contribute more to the performance than that from high-level graphs. The low-level graphs with more nodes can generate more detailed features, and finer features can be better adapted to pixel-level classification tasks.

A certain number of incorrectly predicted pixels are generated under different ablation conditions, as shown in Fig. 12, where fewer red areas indicate better classification performance. In the IN dataset, the most obvious situation occurs after the PLT ablation, which indicates that the PLT contributes

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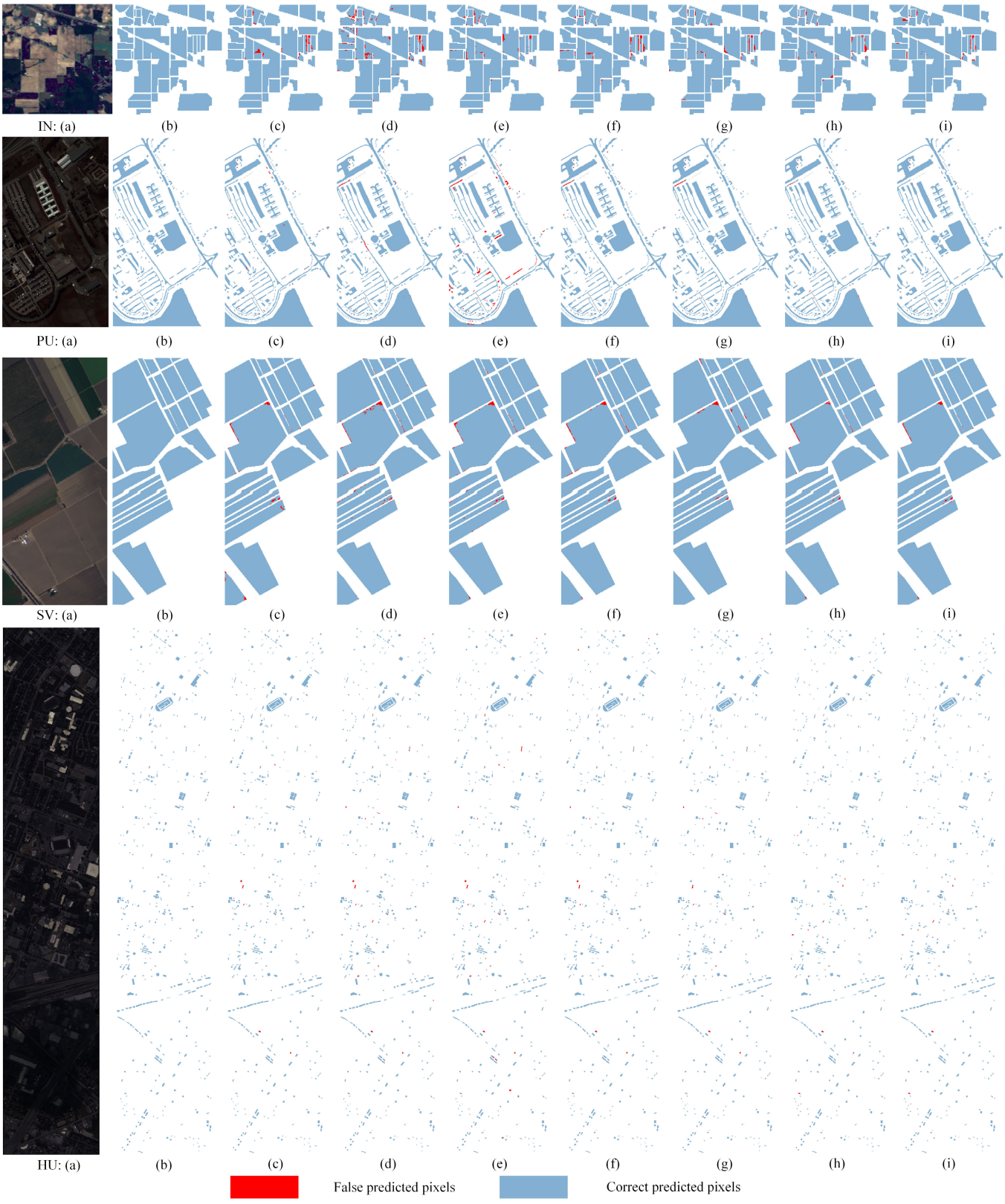


Fig. 12. Predicted pixels under different ablation conditions on four public datasets. (a) False-color image. (b) GT. (c) Without SIA. (d) Without PLT. (e) Without the first skip connection. (f) Without the second skip connection. (g) Without the third skip connection. (h) Without the fourth skip connection. (i) Ours.

the most to the performance improvement on this dataset. Moreover, the first skip connection has the greatest impact on model performance on the PU and HU datasets, owing to many small areas of land coverage in their samples. The detailed information of small areas is lost with the increase in the network stage, which leads to the degradation of classification performance, and the same situation applies to the SV dataset. In general, combining various modules makes the

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TABLE VI
OA (%) OF THE PROPOSED METHOD ON FOUR PUBLIC DATASETS UNDER DIFFERENT MODULE COMBINATIONS

SIA	PLT	S1	S2	S3	S4	IN	PU	SV	HU
○	●	●	●	●	●	98.71±0.13	99.52±0.09	99.46±0.06	98.61±0.18
●	○	●	●	●	●	97.31±0.18	99.37±0.12	99.25±0.05	97.36±0.45
●	●	○	●	●	●	98.71±0.19	97.20±0.18	99.34±0.06	96.25±0.87
●	●	●	○	●	●	98.49±0.11	99.62±0.11	99.46±0.03	98.42±0.33
●	●	●	●	○	●	98.78±0.17	99.66±0.02	99.55±0.09	98.54±0.15
●	●	●	●	●	○	98.87±0.23	99.74+0.05	99.59±0.05	98.73±0.13
○	○	●	●	●	●	97.33±0.15	99.21±0.11	99.14±0.07	96.98±0.24
●	○	●	○	○	○	89.02±0.90	98.22±0.38	98.50±0.49	93.72±2.43
○	●	●	○	○	○	97.38±0.55	99.43±0.10	99.38±0.05	97.17±0.45
●	●	●	●	●	●	**98.98±0.14**	**99.77±0.07**	**99.61±0.07**	**98.80±0.09**

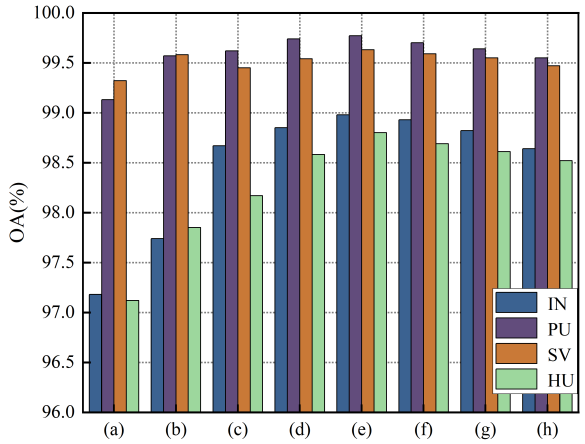


Fig. 13. OA (%) with different concatenate collocations on four public datasets. (a) Series. (b) Concatenation. (c) Parallel of 0.05*SIA and 0.95*SGAT. (d) Parallel of 0.10*SIA and 0.90*SGAT. (e) Parallel of 0.15*SIA and 0.85*SGAT. (f) Parallel of 0.20*SIA and 0.80*SGAT. (g) Parallel of 0.25*SIA and 0.75*SGAT. (h) Parallel of 0.30*SIA and 0.70*SGAT.

network more effective and better identify regional boundary information.

D. Feature Fusion Performance

In order to explore the impact of SIA and SGAT on the accuracy, three collocation forms of these two modules are carried out, that is, connection in series, concatenation, or fusion with different parallel ratios, which are shown in Fig. 13. More precisely, case (a) represents SIA in series after SGAT, case (b) represents a concat operation for the two modules, and then cases (c)–(h), respectively, represent the fusion of the allocated proportion as 0.05, 0.1, 0.15, 0.20, 0.25, and 0.30, which is specified by variable α in (21).

Fig. 13 illustrates that connecting the above two modules in series directly causes information loss because SIA weakens the features extracted by SGAT. Nevertheless, connecting SIA and SGAT modules in (21) can improve network learning performance. With increasing the parallel ratio, the classification performance increases first and then decreases, and expressly, reaches the optimum at $\alpha = 0.15$.

E. Small-Sample Learning Ability

In this section, we explore the learning ability of the MSG-SGAT under finite samples and its advantages over the other methods. The number of training samples of each class in the above four datasets increases with an interval of 5, and the OA of all the comparison methods is plotted in Fig. 14.

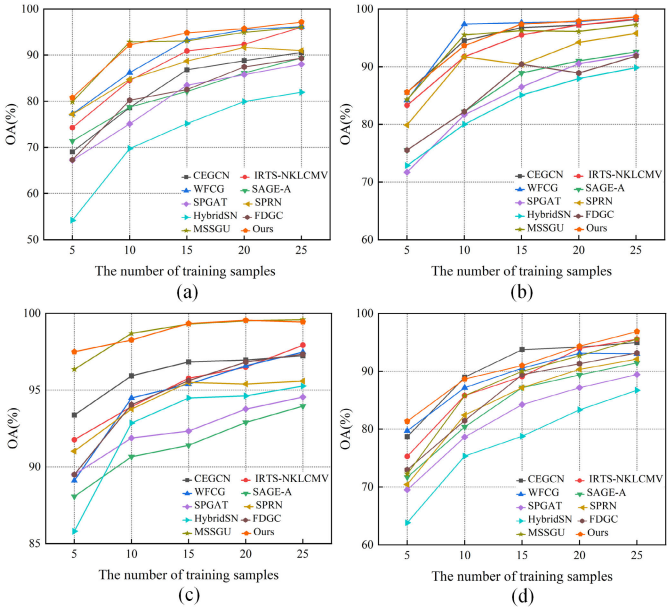


Fig. 14. Small-sample learning ability of each method on four public datasets. (a) IN. (b) PU. (c) SV. (d) HU.

The classification performance of all methods improves greatly with the increase in the training samples, especially the CNN-based methods, which is more dependent on the number of samples. In addition, the OA obtained by HybridSN on each dataset is much lower than others, owing to that the training adequacy and network convergence are limited by the deep structure and sample number. The GNN methods via superpixels, such as CEGCN, WFCG, and MSSGU, always achieve higher accuracy than other CNNs in the case of small-sample training, which verifies the importance of superpixel-guided strategy for classification. It should be noted that the proposed method may not achieve optimal performance across different sample numbers. From Fig. 14(b) and (c), it can be observed that our MSG-SGAT fails to achieve the highest OA at ten training samples, and the same situation occurs when the training number of HU samples is 15. For some datasets, a specific limited number of training samples are not sufficient to cover the entire feature space and capture more complex patterns in the data. The limited generalization ability of the network results in the inability to accurately classify invisible validation samples. However, with the increase in the number of samples, the performance of our method is gradually improved. Furthermore, due to the combination of multiscale spatial and spectral background information at the same time, the OA of our MSG-SGAT is generally better than others on

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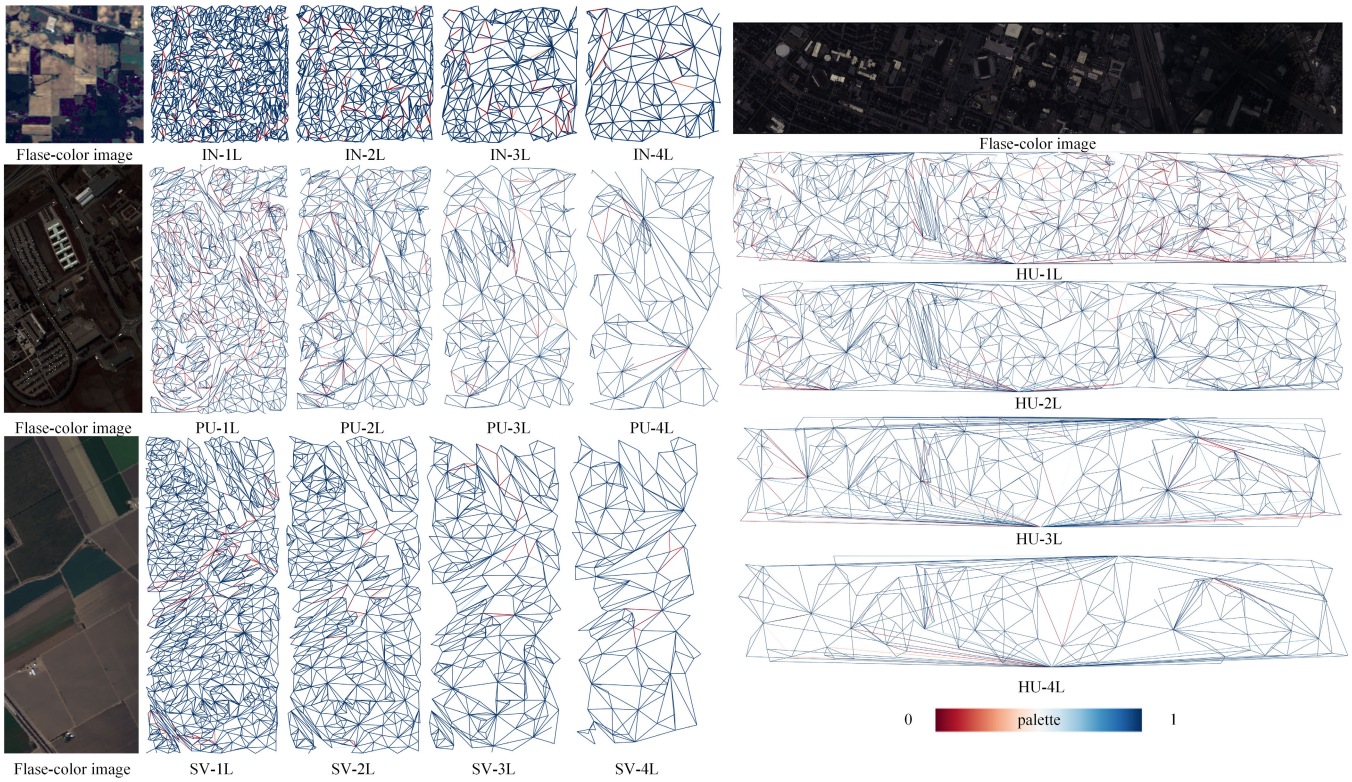


Fig. 15. Edge visualization of learned graphs on the IN, PU, SV, and HU.

the four datasets, and the stability of the validation data is better. The trend in Fig. 14 indicates that our method has better adaptability and effectiveness in dealing with small-sample classification problems.

F. Graph Learning Ability

To further illustrate the effectiveness of the learning ability of our MSG-SGAT, the edge features learned by the encoder on the above four datasets are visualized in Fig. 15, where the edge weights are mapped to [0, 1] and the color palette is set from red to dark blue. The similarity among nodes is represented by different color levels, and the change of color corresponds to the weight size. All the edges of homogenous regions are represented as dark blue, while most of the edges on class boundaries are displayed as red, which indicates that our method has different degrees of information interaction in the homogeneous and heterogeneous regions with excellent specific region discrimination. However, the boundary edges change dark with the deepening of the layers, owing to the aggravation of the oversmoothing problem. The network automatically learns the edge weights of graphs and generates features in different stages from the training samples, and the discrimination ability within and between classes is improved by fusing the features. Although we design appropriate layer numbers and add skip connection to alleviate this phenomenon, the nodes features of different classes become very similar and tend to be homogeneous due to multiple graph convolution.

G. Comparative Experiment of SGAT With GAT

To investigate the accuracy and efficiency of our proposed SGAT module, we conduct experiments to replace the SGAT module in our method with the commonly used GAT and report the results on four datasets in Table VII, which assesses their contribution to the overall network. As listed in Table VII, SGAT can greatly reduce the training time of the network with similar classification accuracy and improve the efficiency by 2∼3 times compared with GAT. We adopt spectral sparring technology in the SGAT module to reduce the calculation burden by trimming some insignificant edges, which benefits from (6) and (7). This innovative approach makes the graph to be sparse and allows the network to process the input data faster, and the test results further demonstrate the high computational efficiency of SGAT. In terms of accuracy, although SGAT slightly reduces the OA value of the network, the difference is not significant. Equation (17) proves that the spectral sparsity of the graph is used to reduce the number of edges, and the characteristics of the computation between the original graph and sparse subgraph are close in

TABLE VII
PERFORMANCE COMPARISON BETWEEN SGAT AND GAT

Dataset	Item	SGAT	GAT
IN	OA(%)	98.98	**99.05**
	Training time(s)	**62.31**	232.53
	Testing time(s)	**0.55**	0.74
PU	OA(%)	99.77	**99.81**
	Training time(s)	**129.52**	302.49
	Testing time(s)	**4.62**	5.76
SV	OA(%)	99.61	**99.65**
	Training time(s)	**103.18**	278.56
	Testing time(s)	**3.11**	3.76
HU	OA(%)	98.80	**98.87**
	Training time(s)	**312.73**	568.60
	Testing time(s)	**15.98**	18.49

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TABLE VIII
INFLUENCE TEST OF INITIAL SUPERPIXEL CONFIGURATION

Dataset	Item	r_s = Initial center number of superpixels : total pixels							
		1/10	1/20	1/50	1/100	1/200	1/500	1/800	1/1200
IN	OA(%)	98.92	**98.98**	98.52	98.14	97.65	96.59	-	-
	Training time(s)	196.57	62.31	30.08	21.89	19.89	17.10	-	-
	Testing time(s)	0.74	0.65	0.60	0.56	0.54	0.52	-	-
PU	OA(%)	-	99.14	99.56	99.67	**99.78**	99.43	99.19	98.81
	Training time(s)	-	5283.16	1076.86	287.21	149.52	112.76	102.32	97.94
	Testing time(s)	-	7.89	6.77	6.15	5.12	4.94	4.65	4.22
SV	OA(%)	99.23	99.51	99.58	**99.68**	99.37	99.05	98.86	98.65
	Training time(s)	3658.31	830.99	246.53	113.18	81.01	74.51	67.92	64.43
	Testing time(s)	4.22	3.64	3.29	3.08	2.93	2.81	2.77	2.71
HU	OA(%)	-	-	97.10	97.84	98.67	**98.80**	98.31	97.95
	Training time(s)	-	-	795.08	556.11	442.63	372.73	347.97	339.24
	Testing time(s)	-	-	19.29	18.54	18.11	17.59	17.21	16.62

the sense of the Frobenius norm. Therefore, SGAT is able to provide classification accuracy comparable to traditional GAT modules, which can provide a more effective and efficient solution for graph-based HSI classification tasks.

H. Initial Superpixel Configuration

For most superpixel segmentation algorithms, the configuration of the initial superpixel center has a direct impact on the subsequent calculation process and results, which is also the case in our superpixel-guided MSG-SGAT. Here, an experiment is performed to analyze the influence of the initial superpixel configuration on the network performance, which is to strike a balance between computation and timeliness, so as to determine the appropriate number of initial centers. In this experiment, we employ a region merging-based superpixel segmentation algorithm to segment samples from four datasets and determine the initial number of superpixels based on different segmentation ratios. Considering the variations in sizes among the samples, fixing the number of initial superpixel centers is insufficient to analyze their guiding effect on the overall network performance. Therefore, here, we take the ratio r_s of the number of superpixels to the total number of pixels in the image as the basis. For instance, for an IN sample with 145*145 pixels, when $r_s = 1/20$, the initial number of superpixels is 1051. As recorded in Table VIII, the network achieves optimal performance on different datasets corresponding to different r_s's. For instance, on the IN dataset, the network achieves the optimal classification performance with $r_s = 1/20$, while, on the PU, SV, and HU datasets, the optimal ratios are 1/200, 1/100, and 1/500, respectively. However, relative to different sample sizes, the optimal classification performance consistently occurs at an initial superpixel size of approximately 1000. Excessive initial superpixel centers will force the separation of ground regions with similar spectral characteristics, that is, the homogeneous region is segmented in a heterogeneous manner, which increases the computational complexity. In addition, these separated homogeneous superpixels may be prone to misclassification during the later merging process due to the loss of details in the feature extraction stage. The experimental results further demonstrate that too many superpixels not only decrease classification accuracy but also increase time overhead. On the other hand, too few initial centers will result in a single superpixel potentially containing heterogeneous regions with different spectra, which cannot adequately represent the details and local features in the image, resulting in reduced accuracy and boundary confusion.

Our MSG-SGAT is a multistage network, and the transfer of intermediate nodes in each stage is guided by the merging strategy described in Section III. For small-size samples, selecting a too large value of r_s, that is, too small an initial number of centers, will cause the superpixels in the deep structure to be rough due to a large number of heterogeneous regions and, thus, cannot accurately assign the superpixels to the correct category. Therefore, we use "−" on the scale of 1/1200 for the IN dataset, indicating that the results are not comparable. For larger samples, such as PU and HU, selecting a smaller r_s will lead to significant computational overhead and noise sensitivity issues for each layer in the multilevel graph structure, and the superpixel guidance will lose its significance, which is described with "−" in $r_s = 1/10$ and 1/20. In this experiment, we determine the appropriate range of the initial superpixel numbers to achieve the best performance in the superpixel-guided algorithm while weighing computational complexity and timeliness.

V. CONCLUSION

Aiming at solving the problems in the application of HSI classification, such as noise interference, band redundancy, and insufficient sample, we propose the MSG-SGAT. The network constructs a multilevel graph structure guided by superpixels, which can achieve end-to-end training with high accuracy and inference efficiency. A superpixel guidance strategy is adopted to obtain the adjacency and correlation matrices at different stages. In addition, we design an SGAT module to reduce the computational burden, so as to make the network lightweight. SGAT trims the original graph by spectral sparsification to generate a random subgraph, which retains the original spectral information. The features calculated by SGAT between the random subgraph and the original graph only produce small errors, which is proven theoretically and manifests the rationality of our proposed module. While maintaining its accuracy advantage, SGAT simplifies the architecture of GATs, reduces the risk of overfitting, and improves the robustness of graph node classification. Moreover, in each stage of the encoder and the decoder, the features of spatial and channel dimensions are aggregated by concatenating the SGAT and

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SIA branches. The correlation matrix is employed to realize the characteristic transformation between the adjacent layers, and finally, the pixel-level features are further refined to improve classification accuracy. A series of comparison and ablation experiments are conducted on four real HSI datasets, which shows that the proposed MSG-SGAT outperforms other CNN and GNN-based HSI classification methods.

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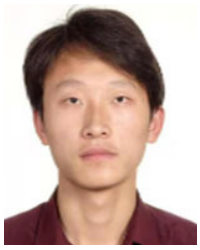
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